

STAR WARS
EPISODE I
THE FILTER MENACE

CuttingGarden 2026 – Würzburg

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CNRS Postdoctoral
Research Fellow

Plan

- Definition and Motivation:
 - What's a signal or noise?
 - Noise in EEG
- What's a filter?
 - Definition
 - Intuition behind dot product and convolution: the foundation of filtering!
 - Causal and Acausal filters limitations
- Danger zone
 - Impact of filter on standard signal (rule of thumbs on what to expect)
 - The hidden costs: phase distortion, edge artifacts, temporal smearing, spurious oscillations, etc.
- How to fix it?
 - Low-pass, high-pass, band-pass, etc.
- Conclusion and Practical guidelines

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- *Filters: When, Why, and How (Not) to Use Them*, Alain **de Cheveigné** and Israel **Nelken**, Neuron, 2019
- *Digital filter design for electrophysiological data – a practical approach*, Andreas **Widmann**, Erich **Schröger**, Burkhard **Maess**, J. Neurosci. Methods., 2015
- Chapter 10 of *Analyzing Neural Time Series Data: Theory and Practice*, Mike X **Cohen**, 2014
- *Four common conceptual fallacies in mapping the time course of recognition*, Rufin **VanRullen**, Front. Psychol., 2011
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Some general insight on signal and noise...

A *signal* is the meaningful information or desired physical measurement you want to detect or transmit, while *noise* is the random, unwanted fluctuation or disturbance that interferes with it.

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As a telecommunications engineer:

1. Signal: human voice
2. Noise: bird chipping

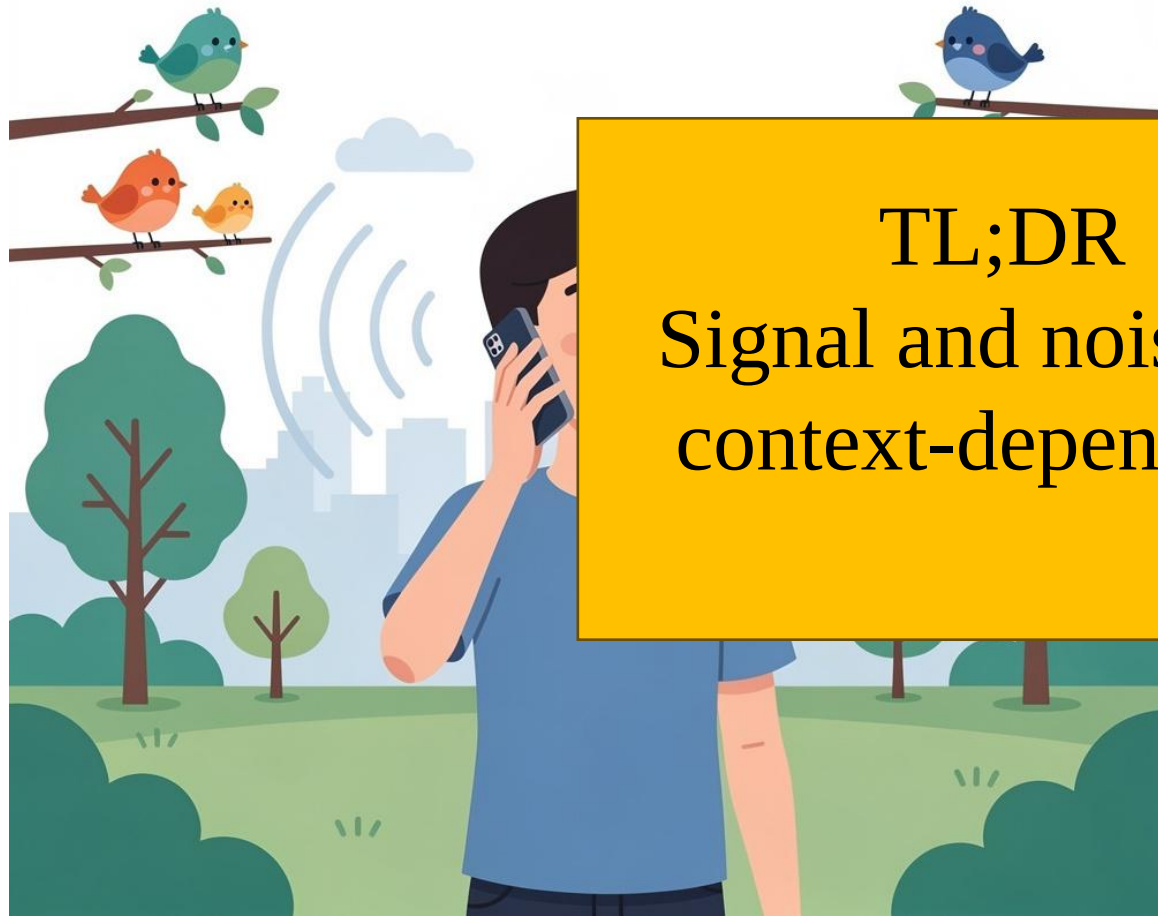
But, as an ornithologist:

Signal: bird chirping

Noise: human voice.

Some general insight on signal and noise...

A *signal* is the meaningful information or desired physical measurement you want to detect or transmit, while *noise* is the random, unwanted fluctuation or disturbance that interferes with it.



TL;DR
Signal and noise are
context-dependent.

communications engineer:

human voice

bird chipping

ornithologist:

bird chirping

noise: human voice.

Noise in EEG

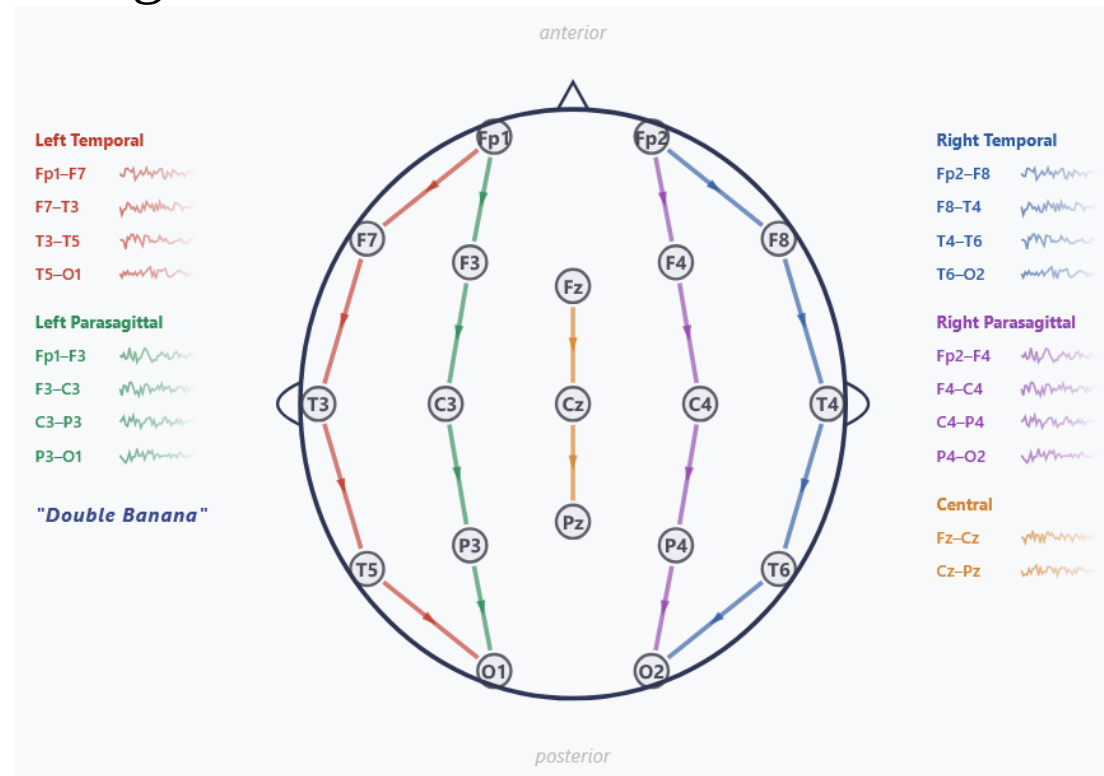
In EEG, we have many types of noise:

- Environmental noise: e.g. 50/60 Hz power line
- Instrument noise: e.g. poor electrode contact, high impedance
- Physiological noise: e.g. ECG-related artifact

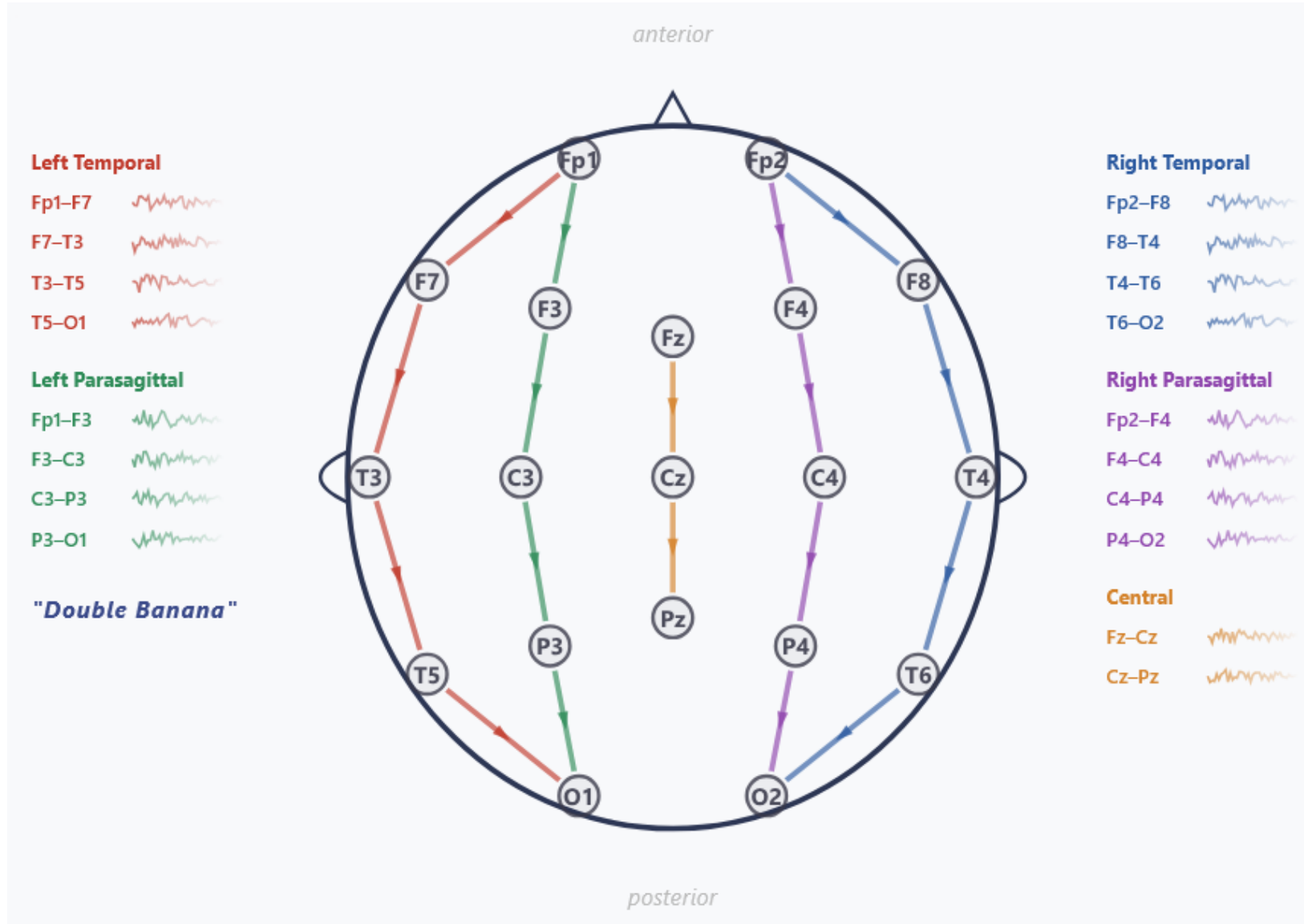
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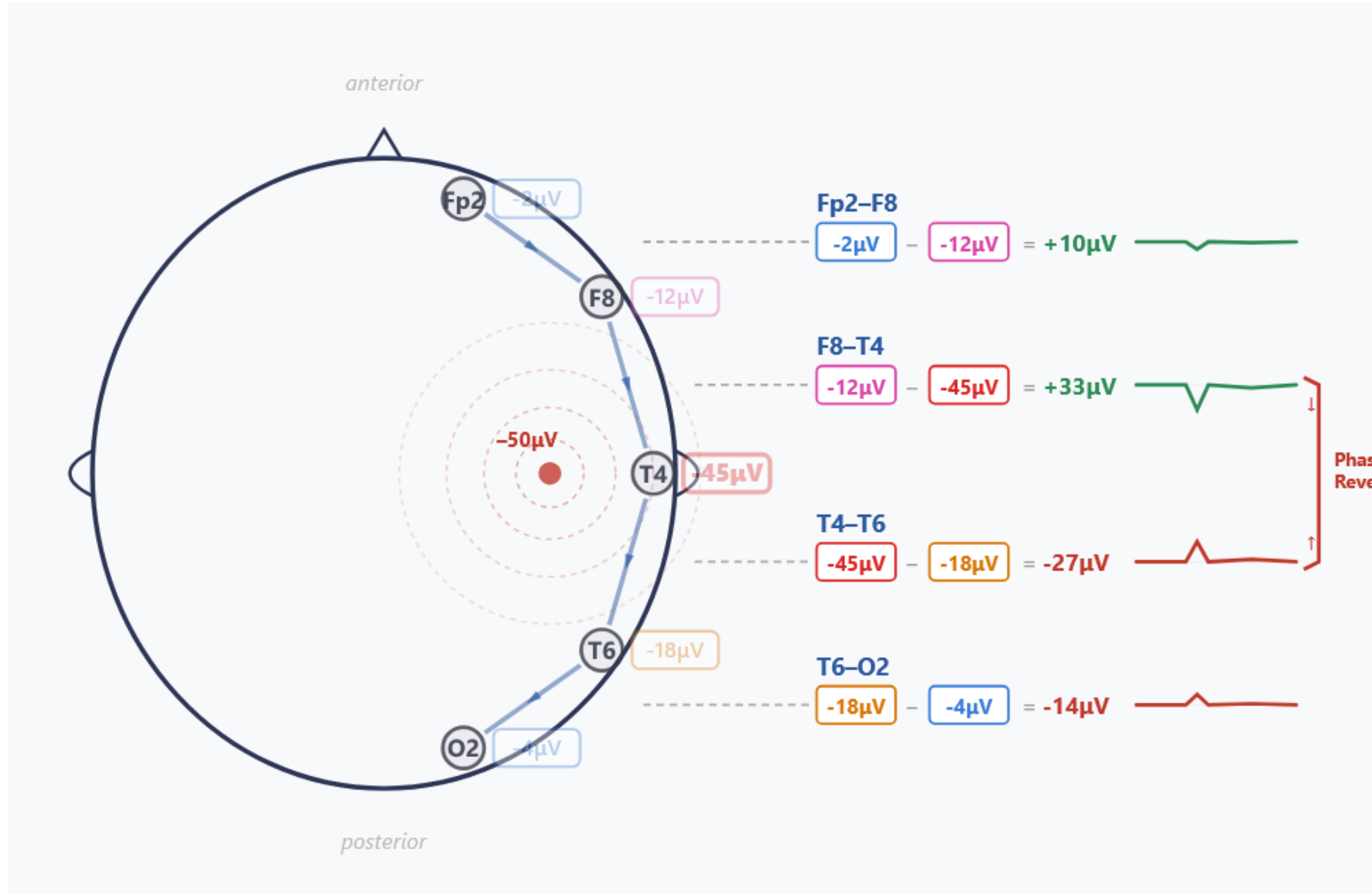
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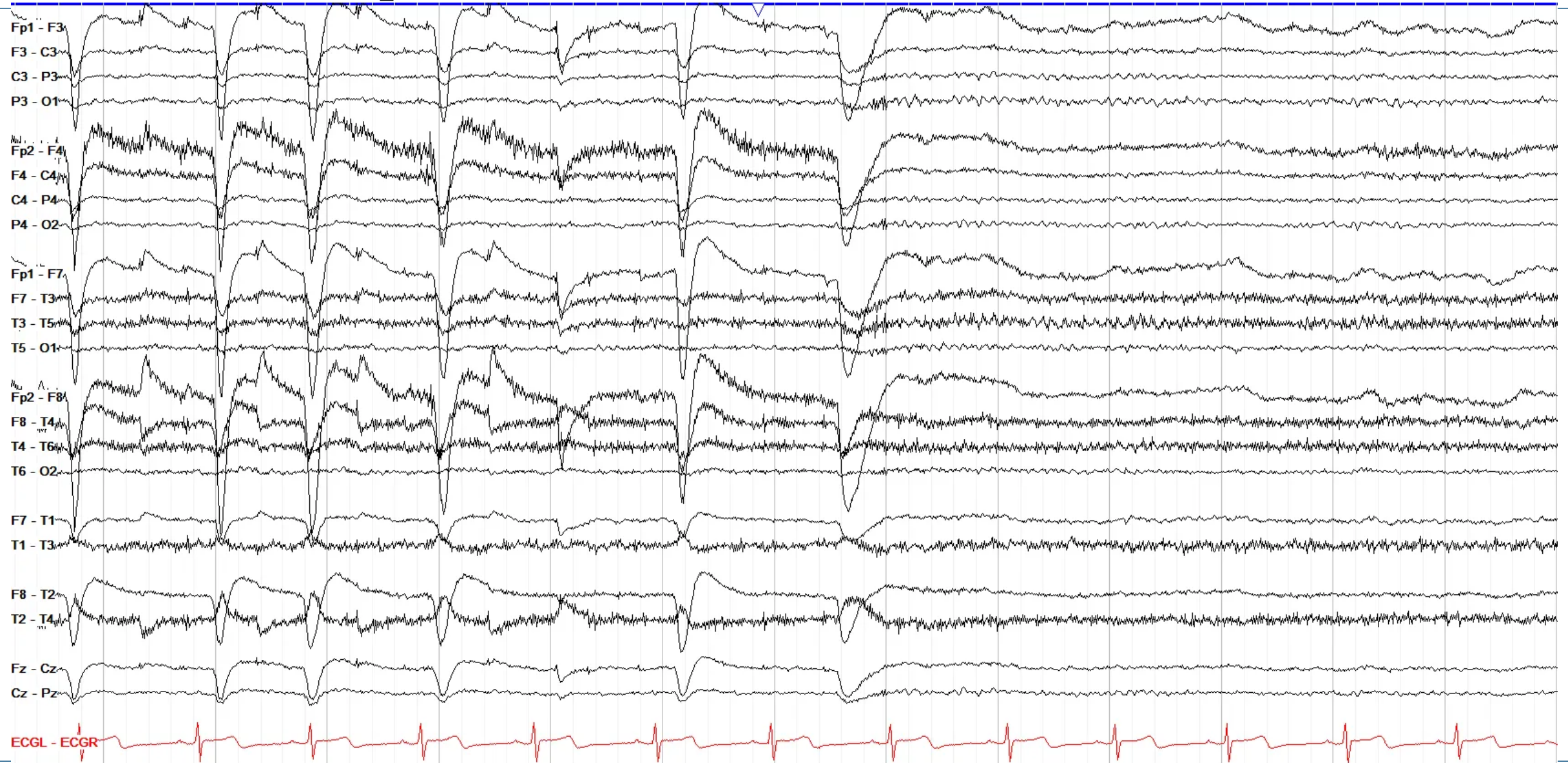
“Double-banana” montage



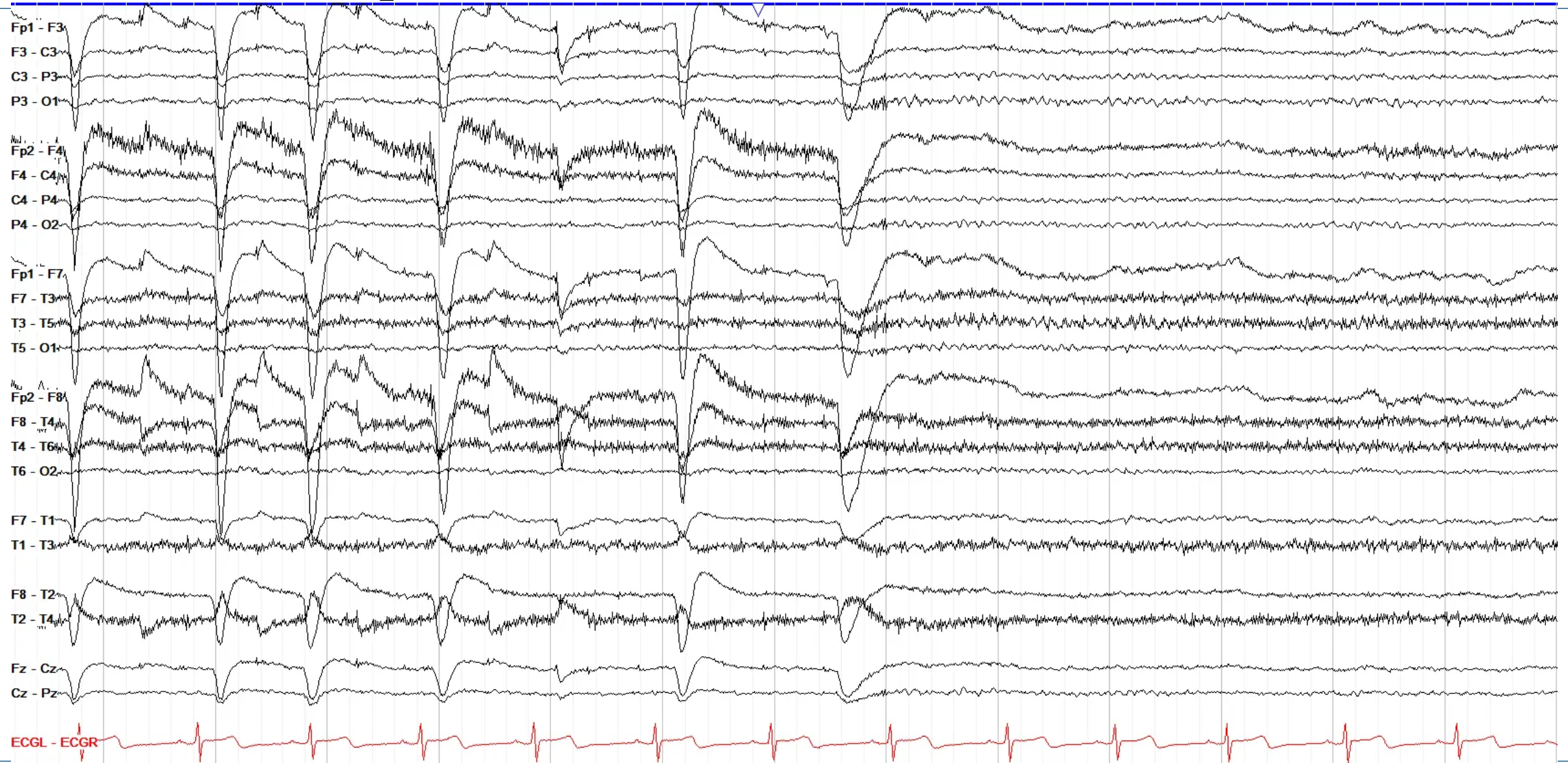
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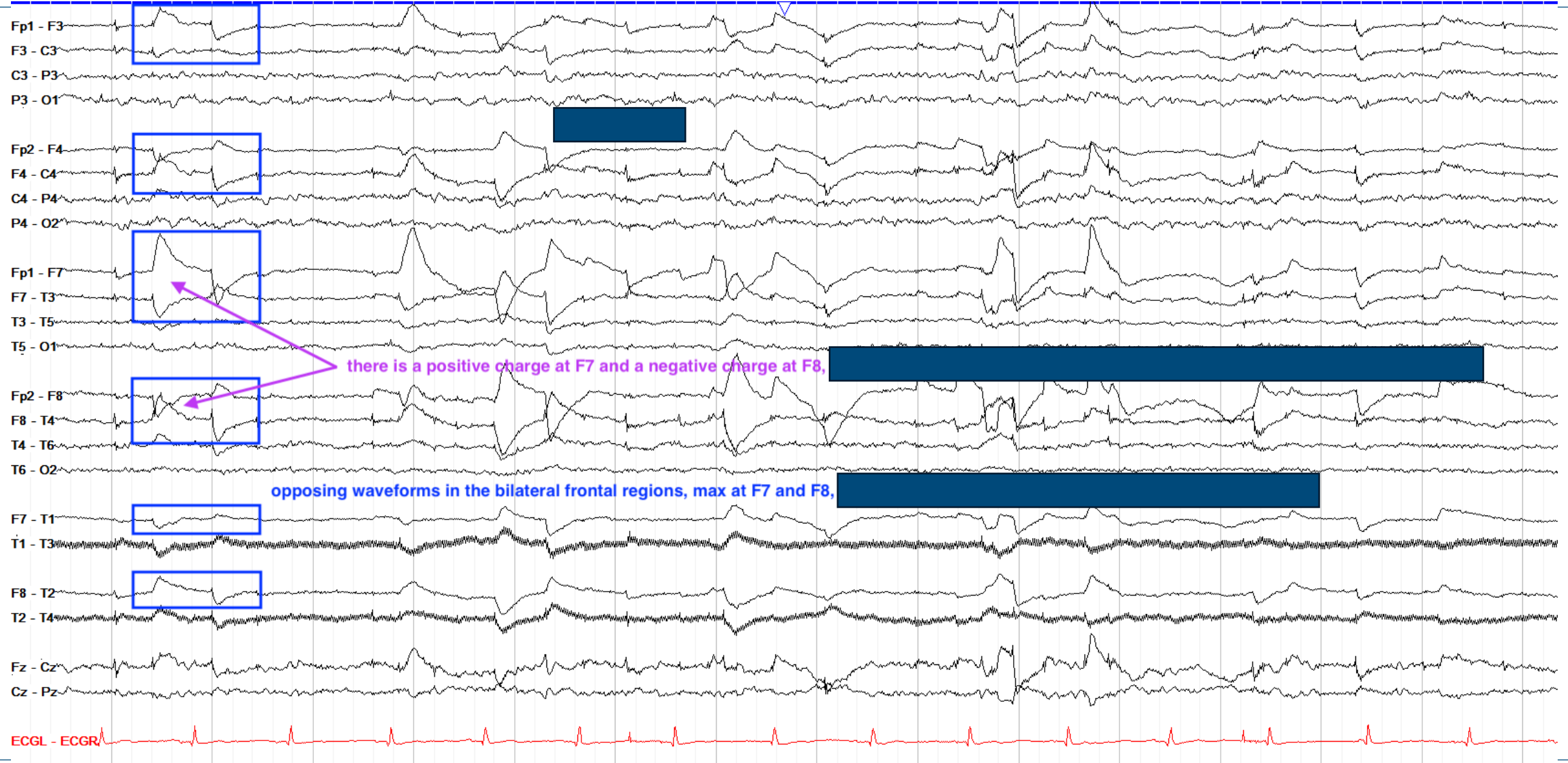
Noise example 1



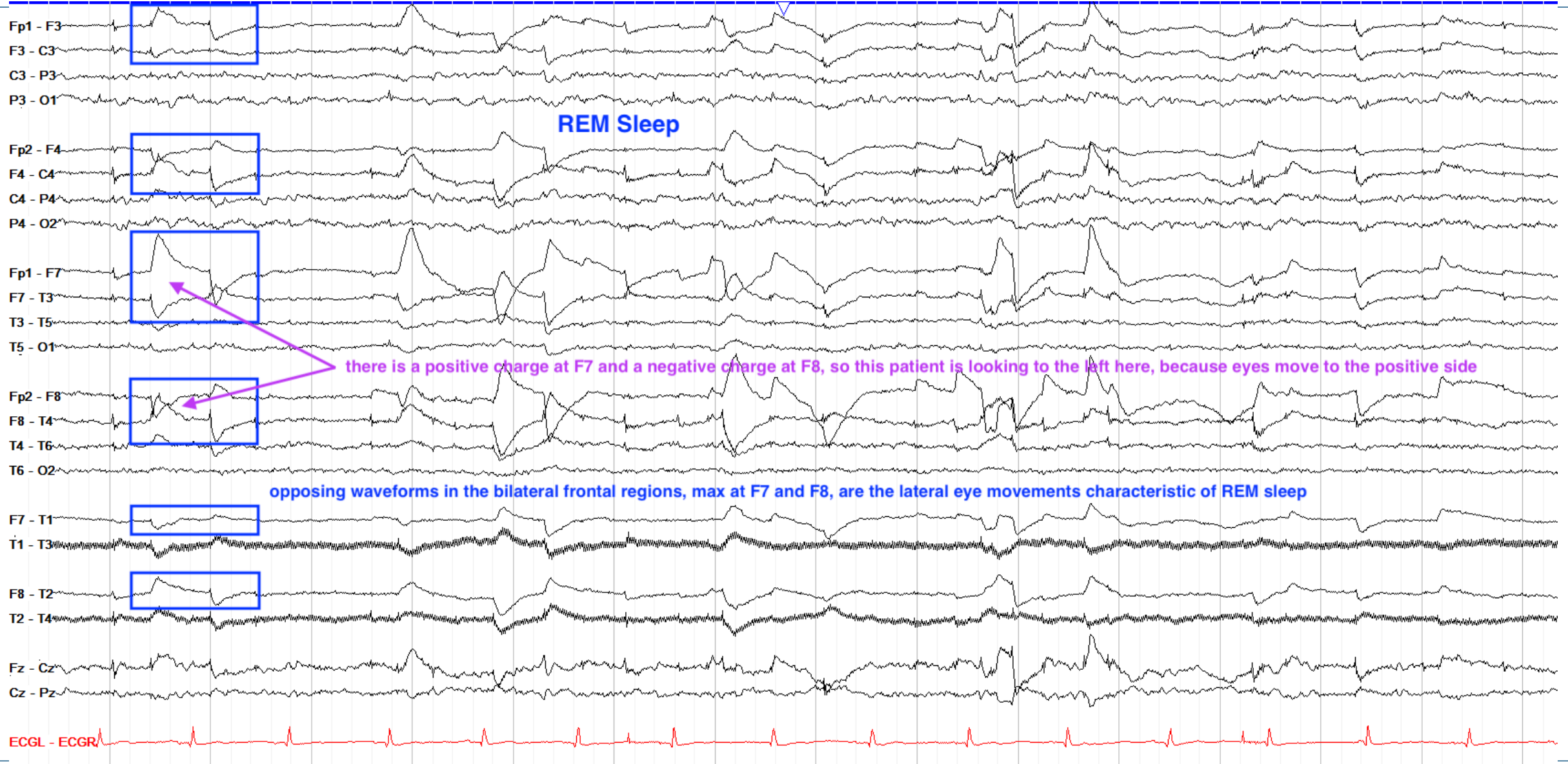
Noise example 1: Blinks



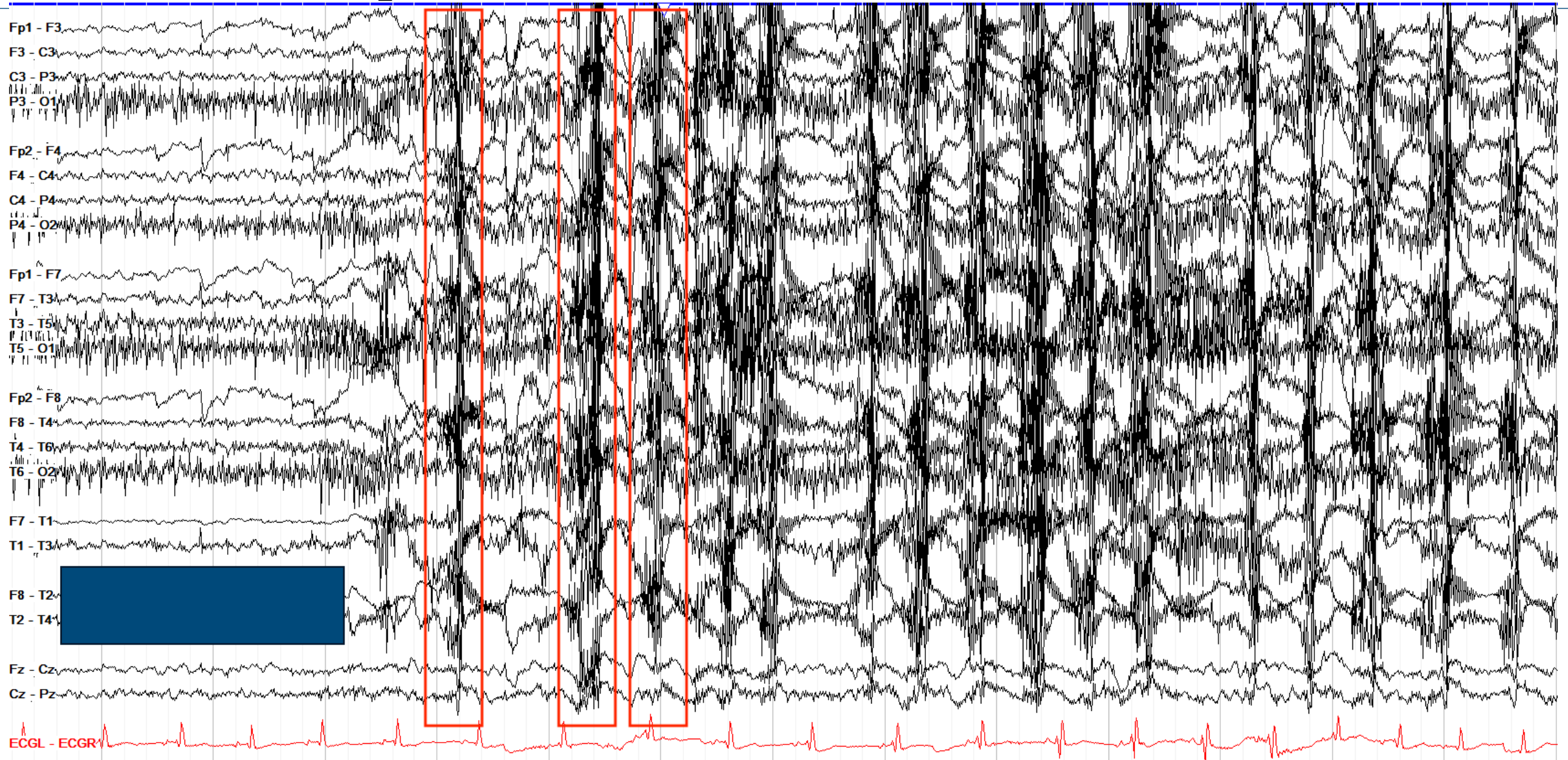
Noise example 2



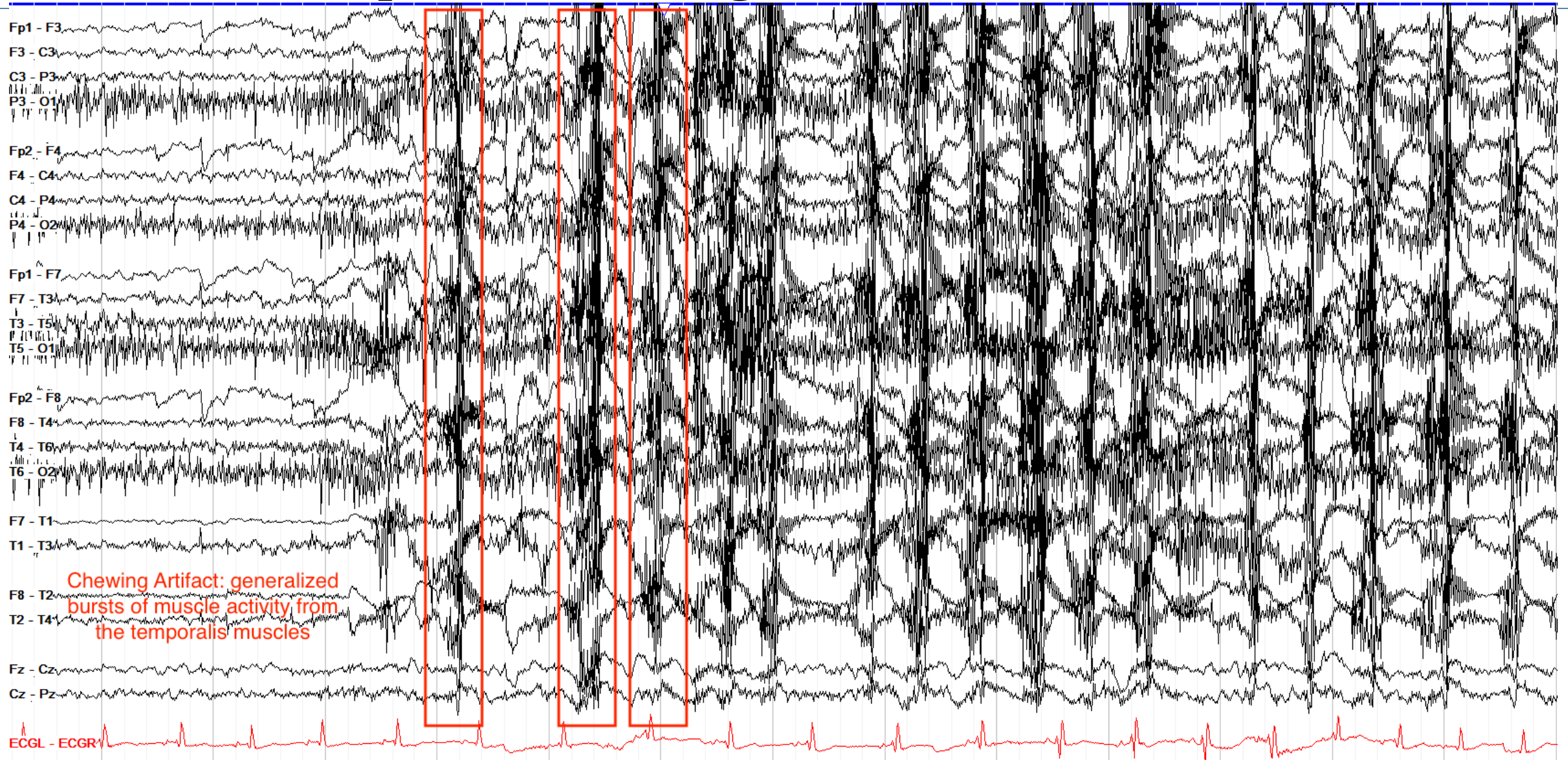
Noise example 2: Lateral eye movements



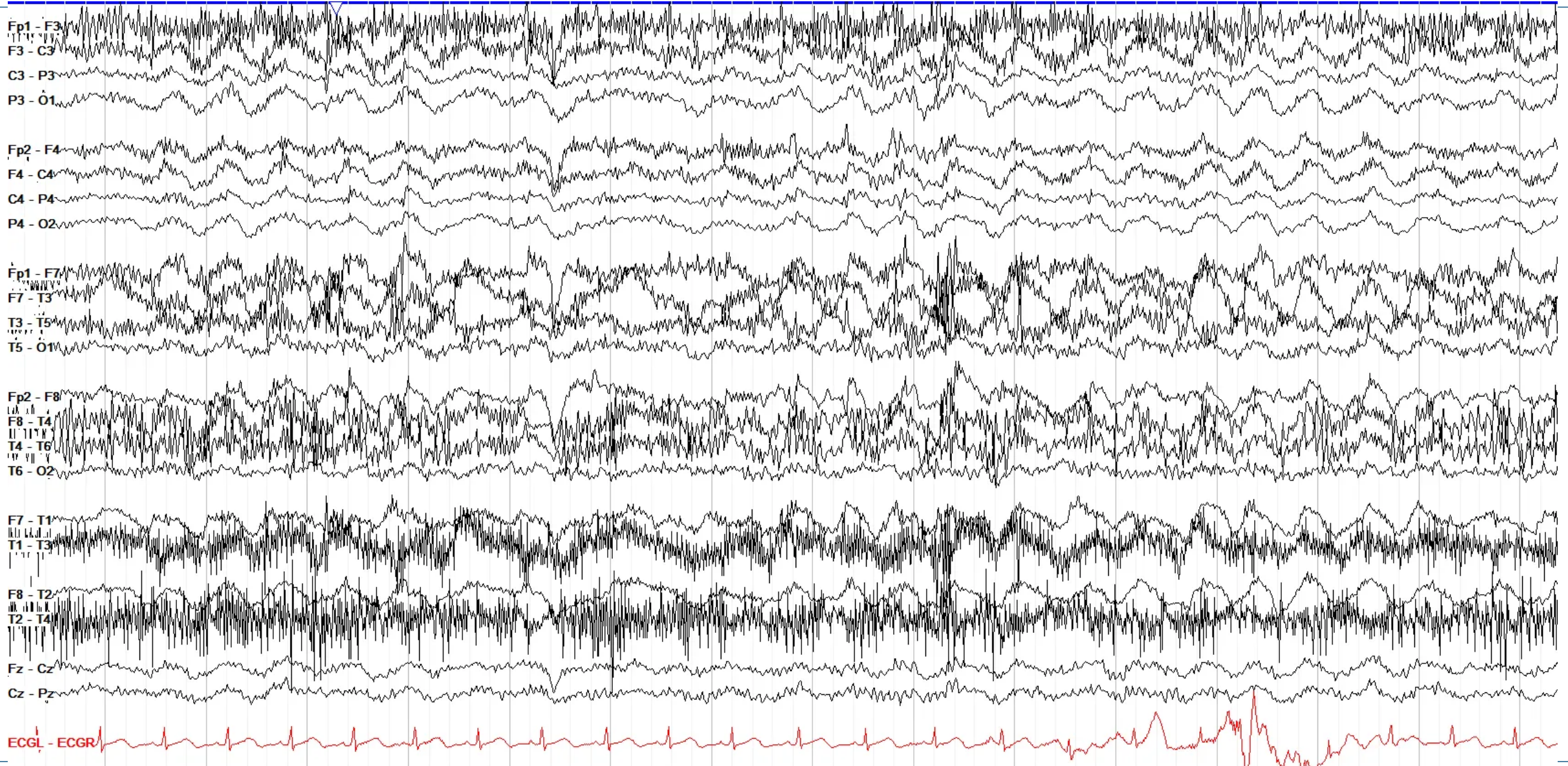
Noise example 3



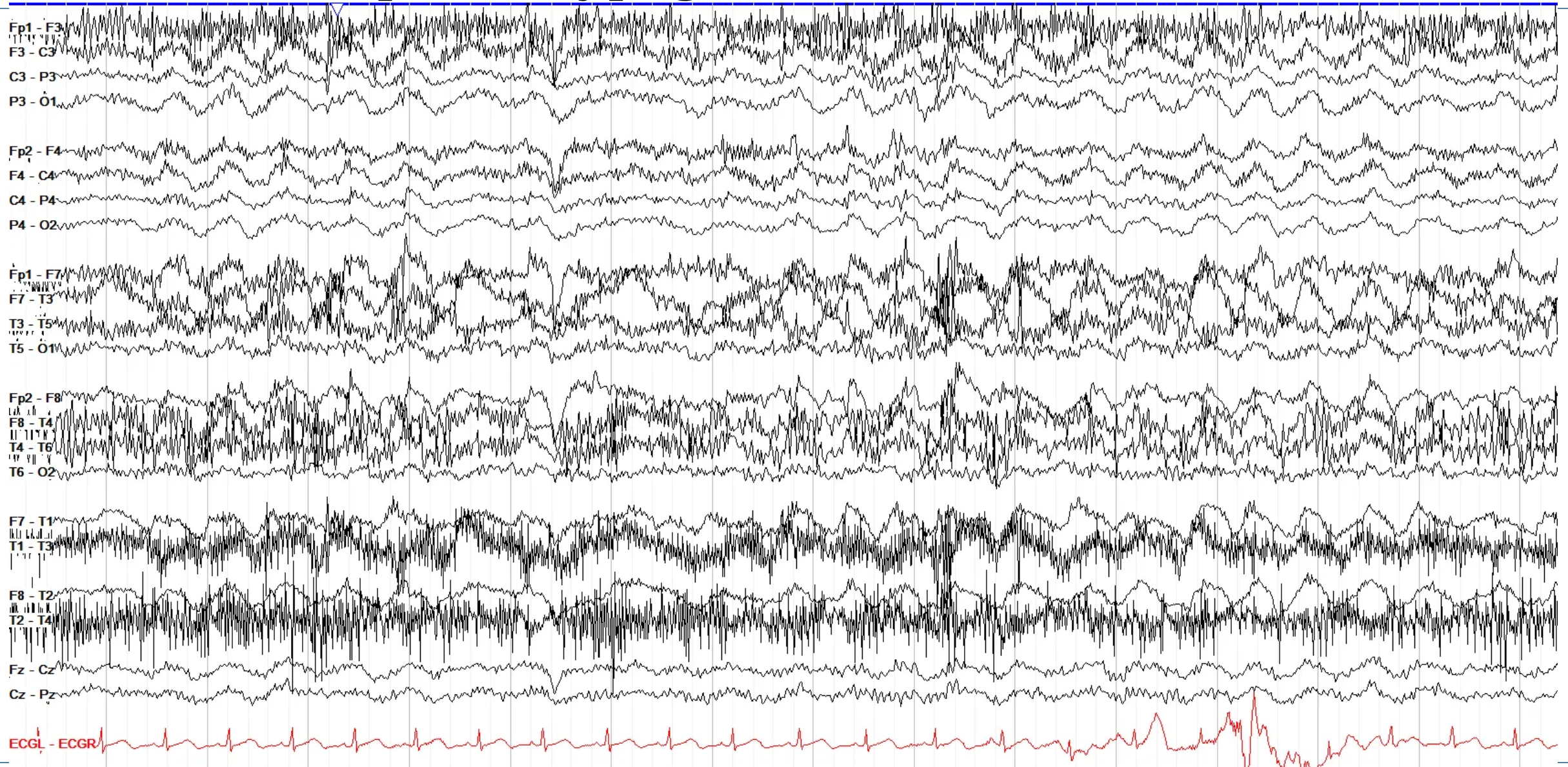
Noise example 3: Chewing



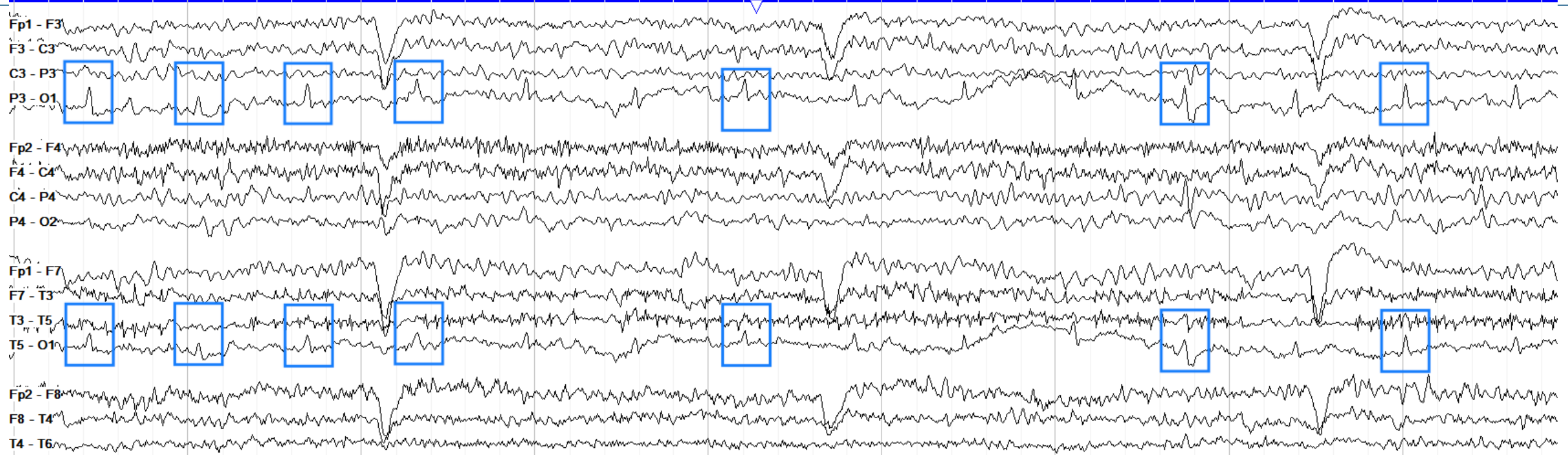
Noise example 4



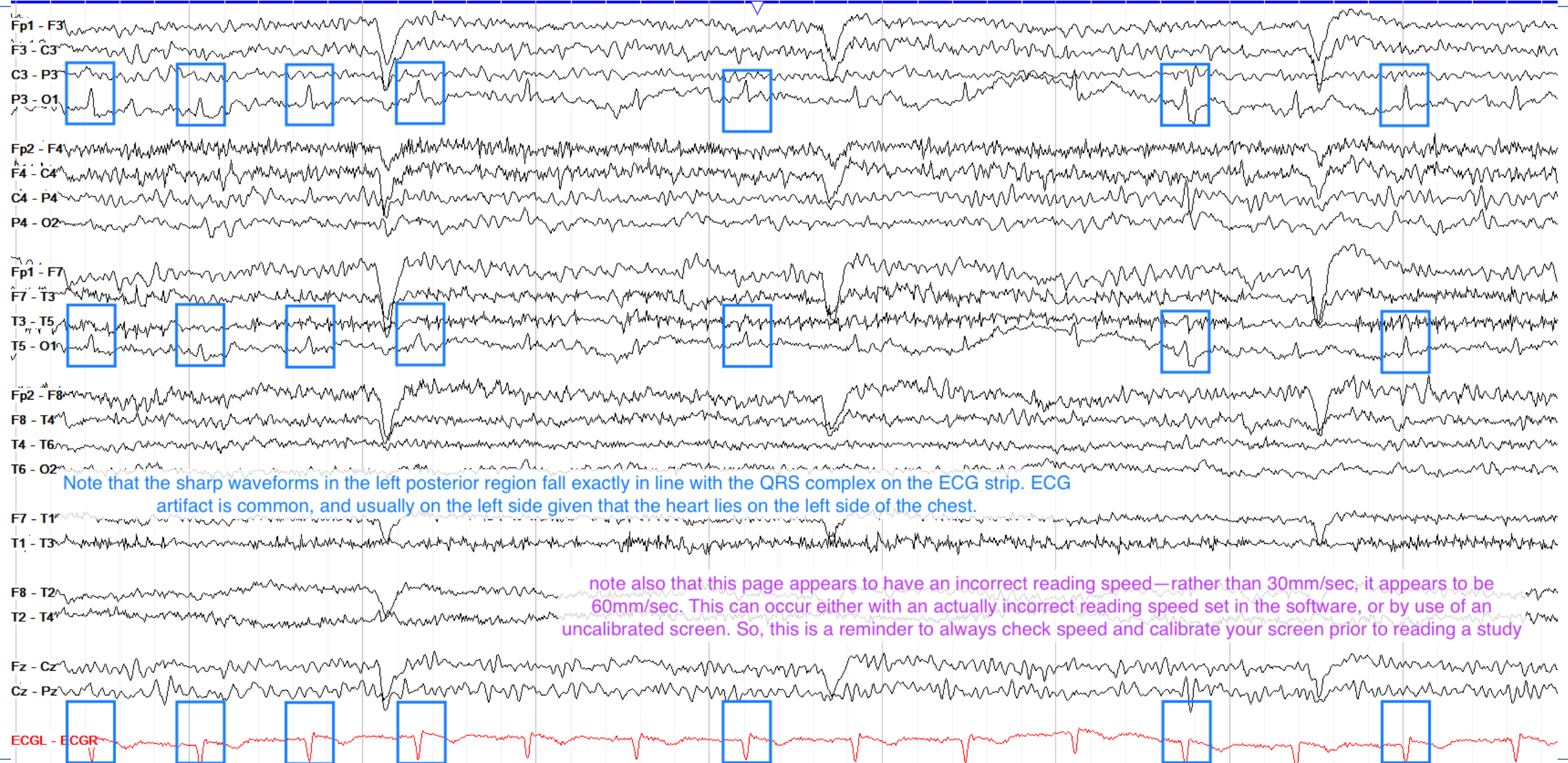
Noise example 4: Hypoglossal movement



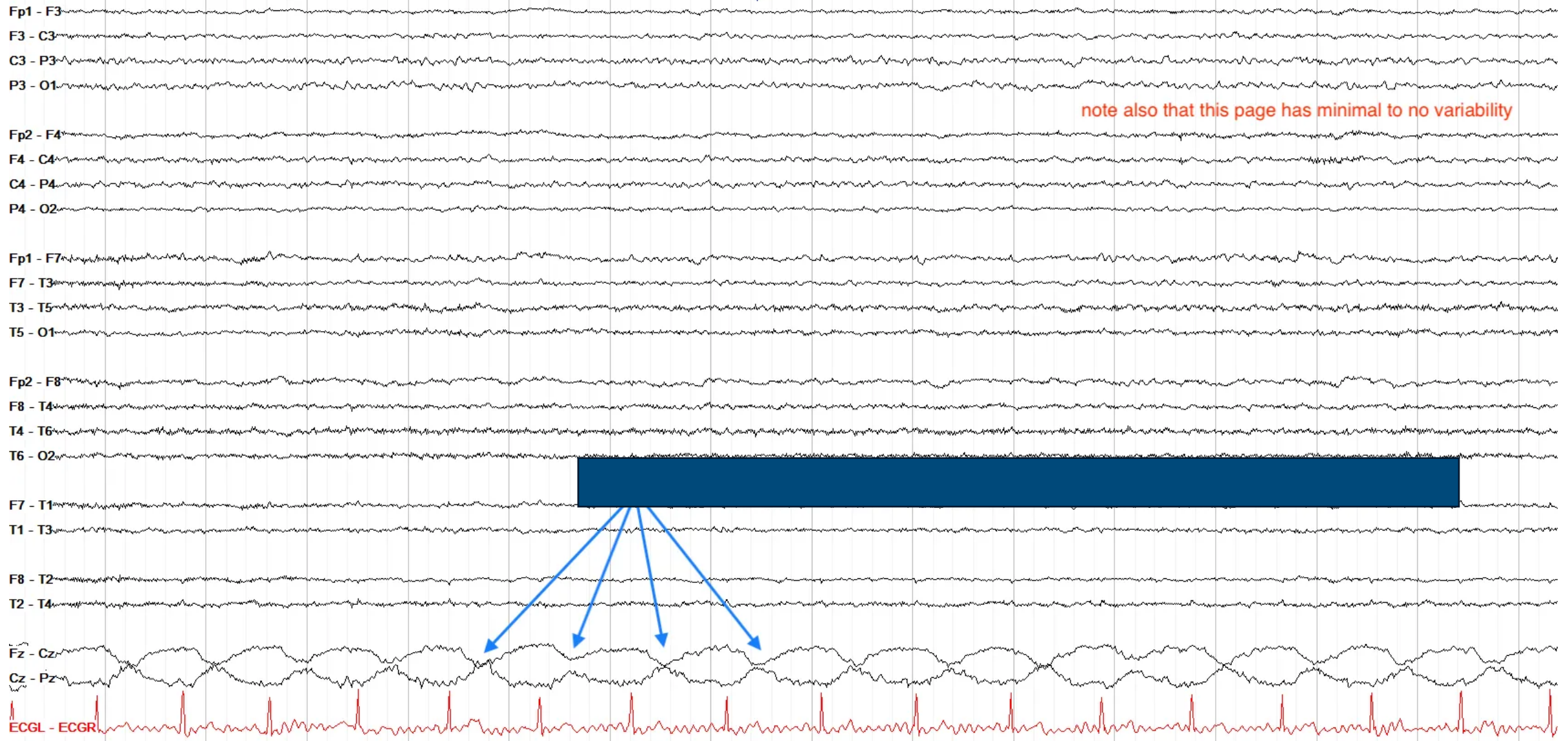
Noise example 5



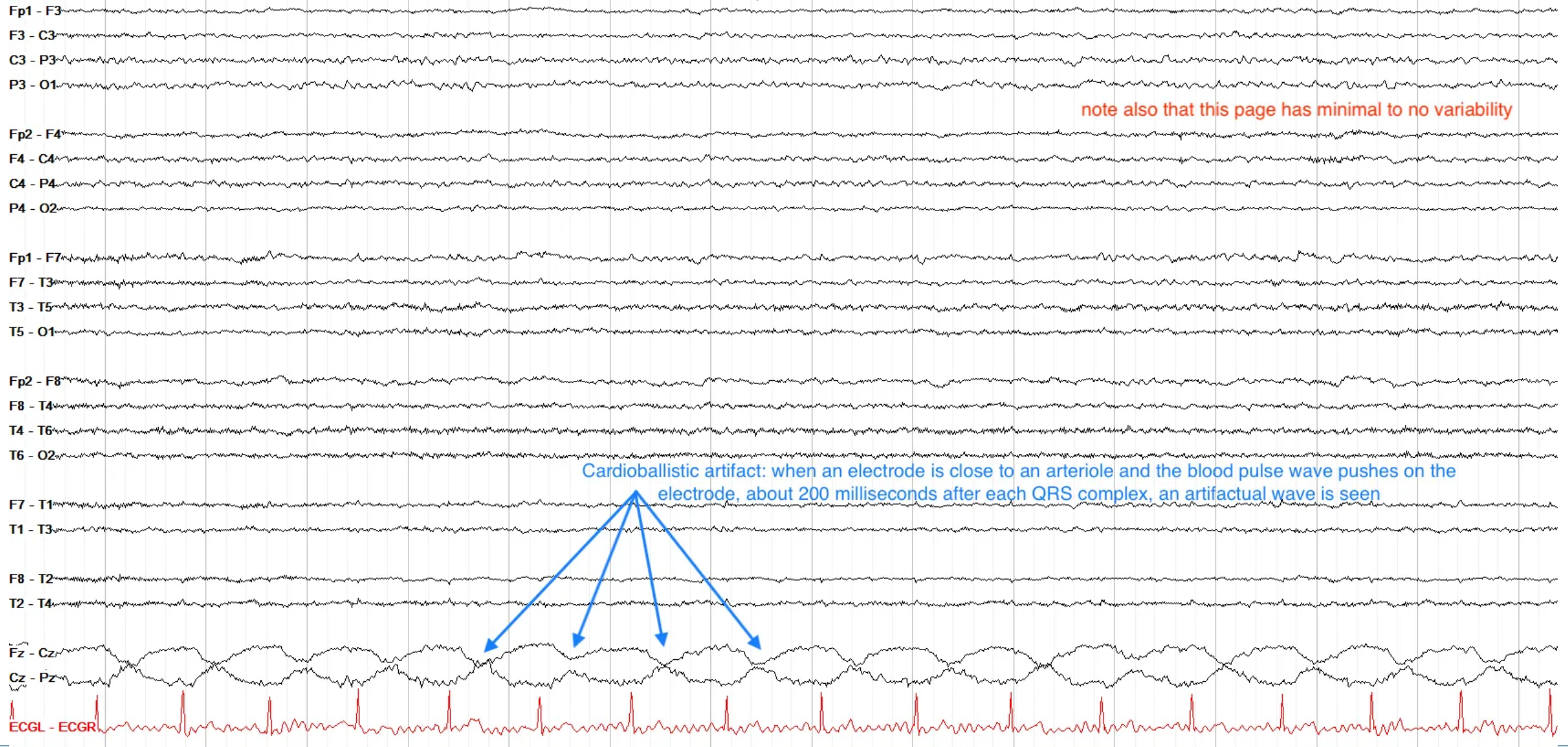
Noise example 5: ECG



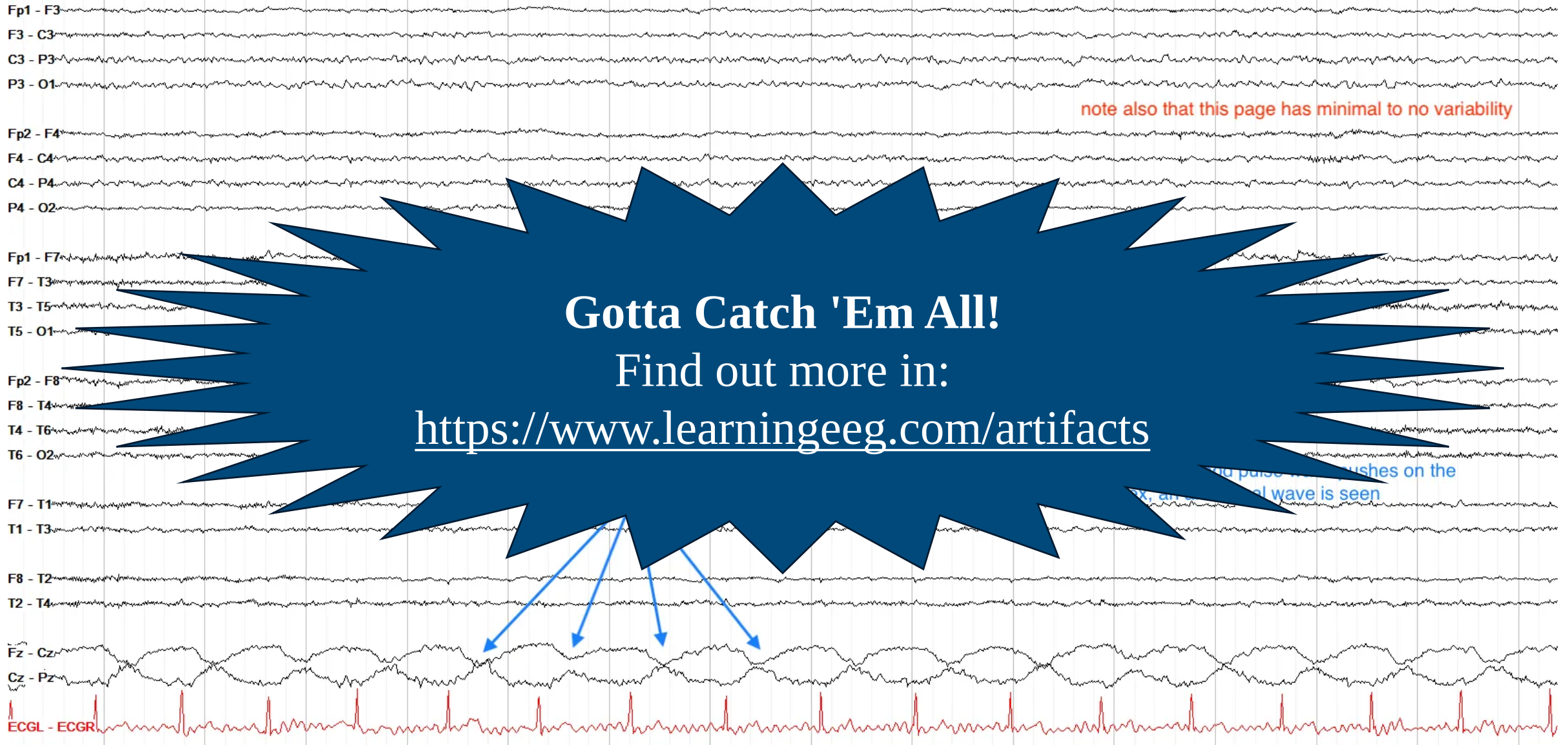
Noise example 6



Noise example 6: Cardioballistic artifact



Noise example 6



Filters can help mitigate such noise, but...

Signal and noise must occupy different spectral regions.

Typically, not true for blinks:
=> Other tools, e.g. ICA

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What's a filter?

“ A filter is an operation that produces each sample of the output waveform y as a weighted sum of several samples of the input waveform x .” (de Cheveigné et al. 2019)

For digital filter (causal):

$$y(t) = \sum_{n=0}^N h(n)x(t - n),$$

where t is the analysis point in time and $h(n), n = 0, \dots, N$ being the impulse response of the filter. This operation is called a *convolution*.

Dot product

For two vectors, $\vec{a}$ and $\vec{b}$, the dot product is defined as:

$$y(t) = \sum_{i=0}^N a_i b_i .$$

Several possible interpretation:

1. It's the sum of the elements from one vector weighted by the element of another vector. (signal-processing interpretation)
2. It's a covariance calculation (statistical interpretation):

$$Cov(\vec{a}, \vec{b}) = \frac{1}{n-1} \sum (a_i - \bar{a})(b_i - \bar{b}).$$

3. A mapping between vectors. (a geometric interpretation)

Convolution, in the time domain, is an extension of the dot product, in which the dot product is computed repeatedly over time.

Interpretation of the convolution operator

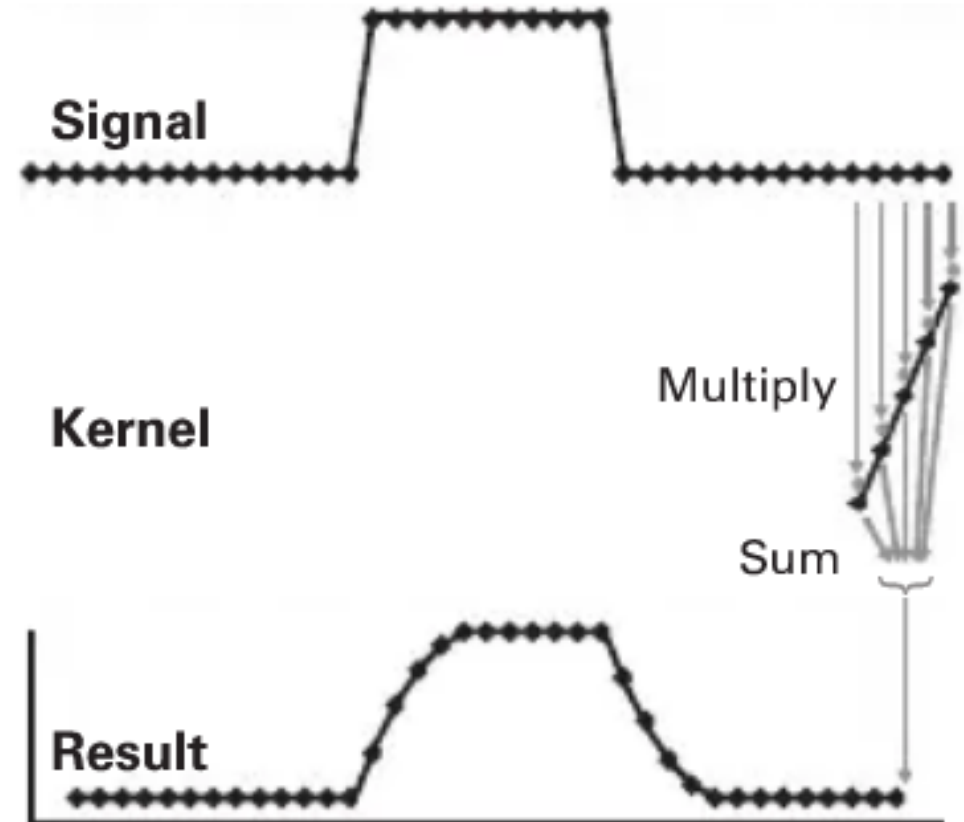
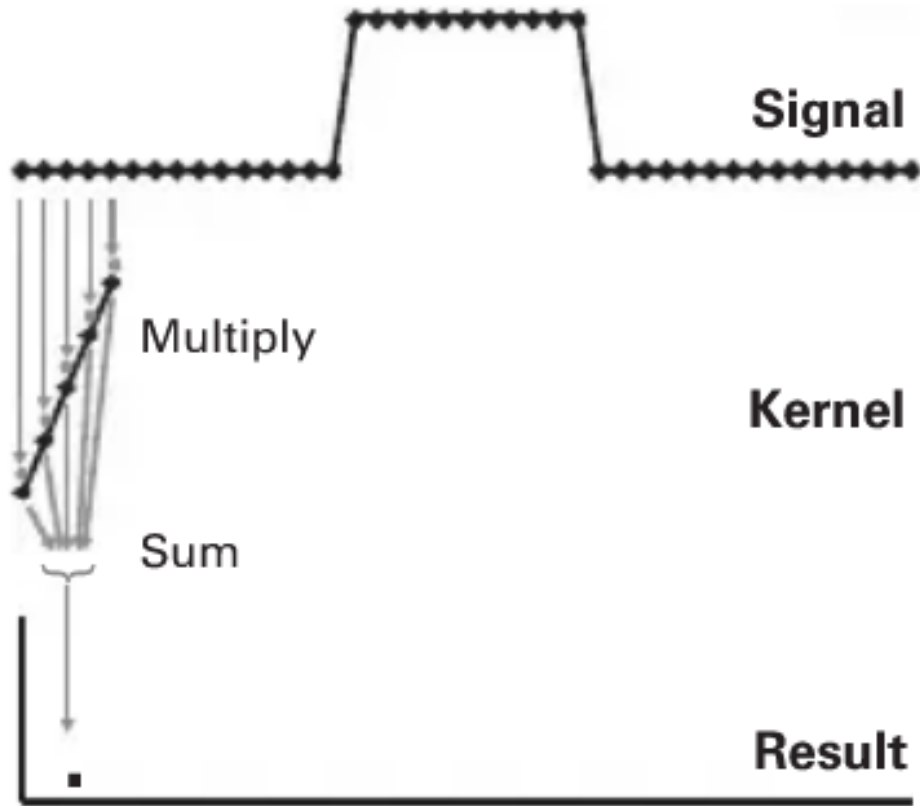
Convolution of x and h :

$$y(t) = \sum_{n=0}^N h(n)x(t - n) .$$

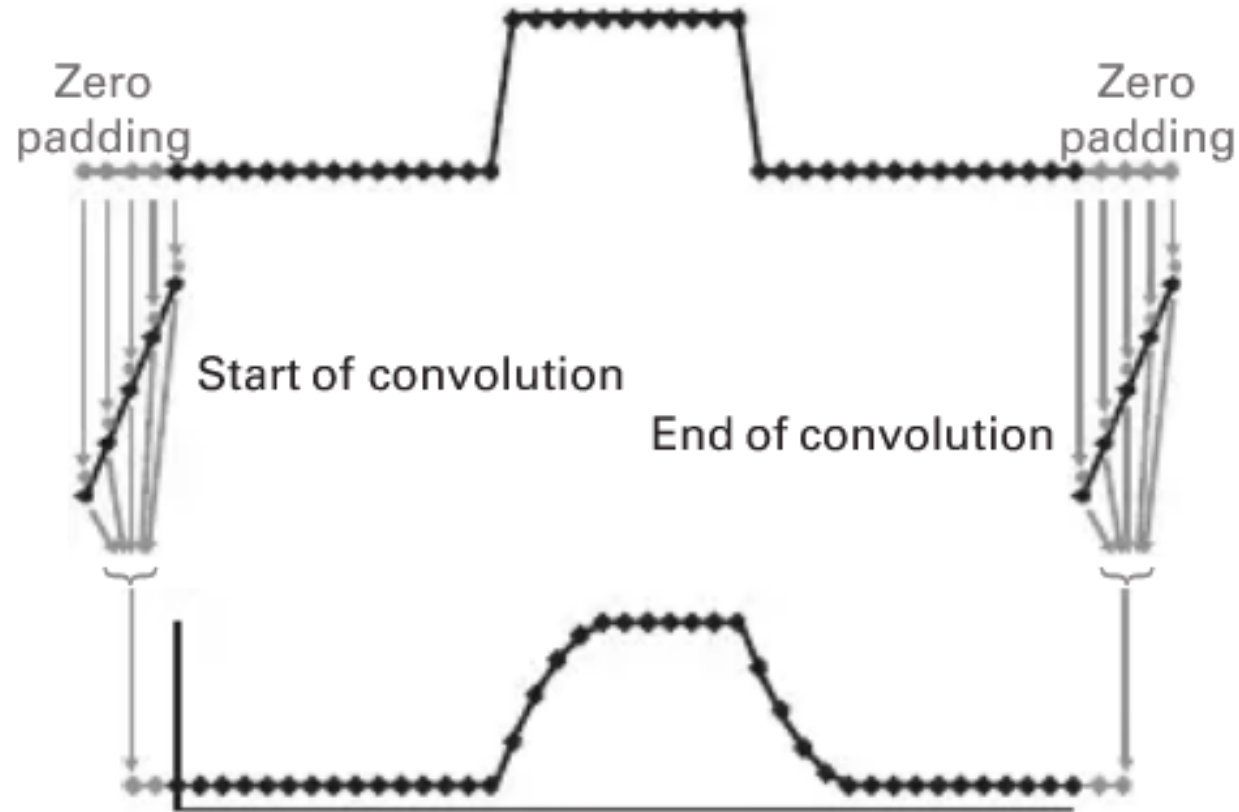
1. It's a time series of one signal weighted by another signal that slides along the first signal (a signal-processing interpretation)
2. It's a calculation of the similarity between two vectors over time: a cross-covariance (a statistical interpretation)
3. It's a time series of mappings between two vectors (a geometric interpretation)

For convenience, x is labelled the *signal* and h is called the *kernel*.

Convolution toy example

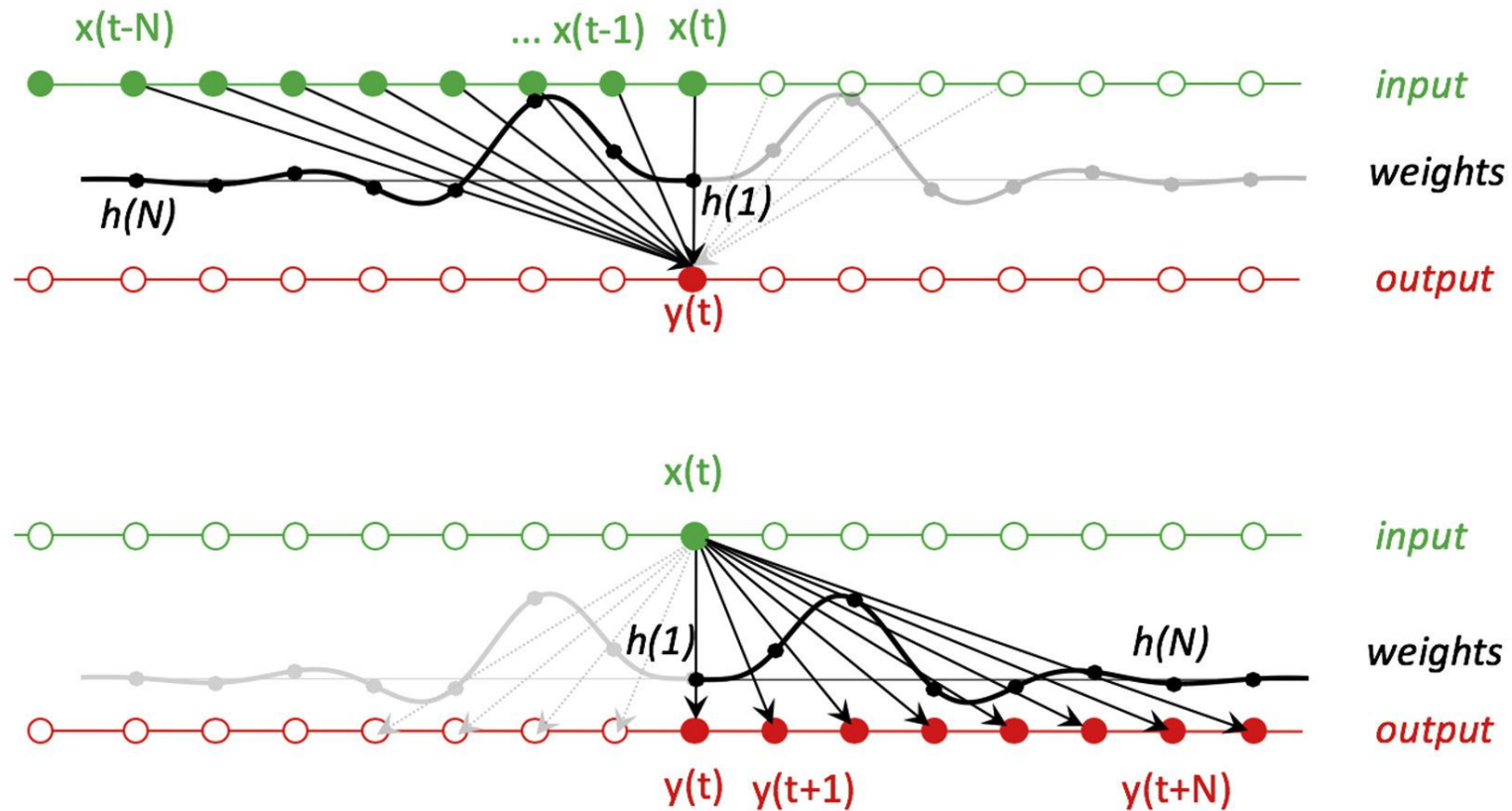


Convolution toy example: zero padding

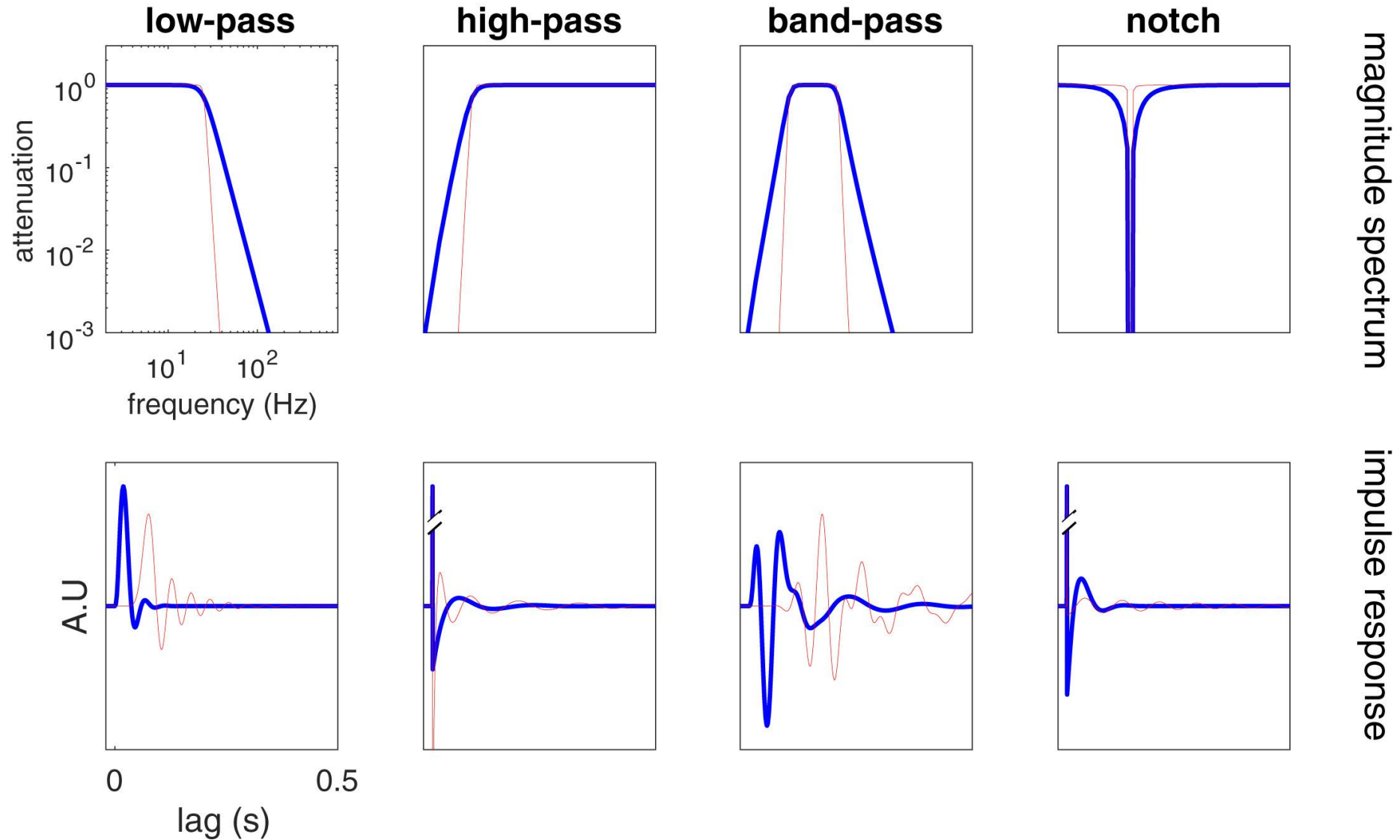


TL;DR: Filtering

Every sample of the output depends on multiple samples of the input.

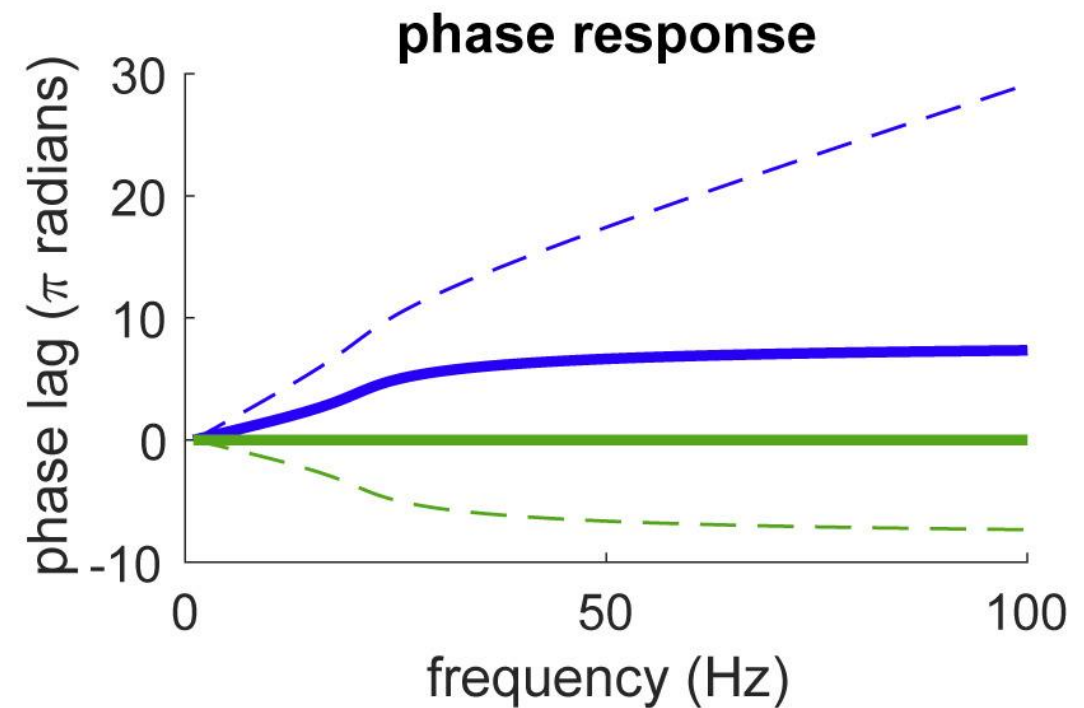
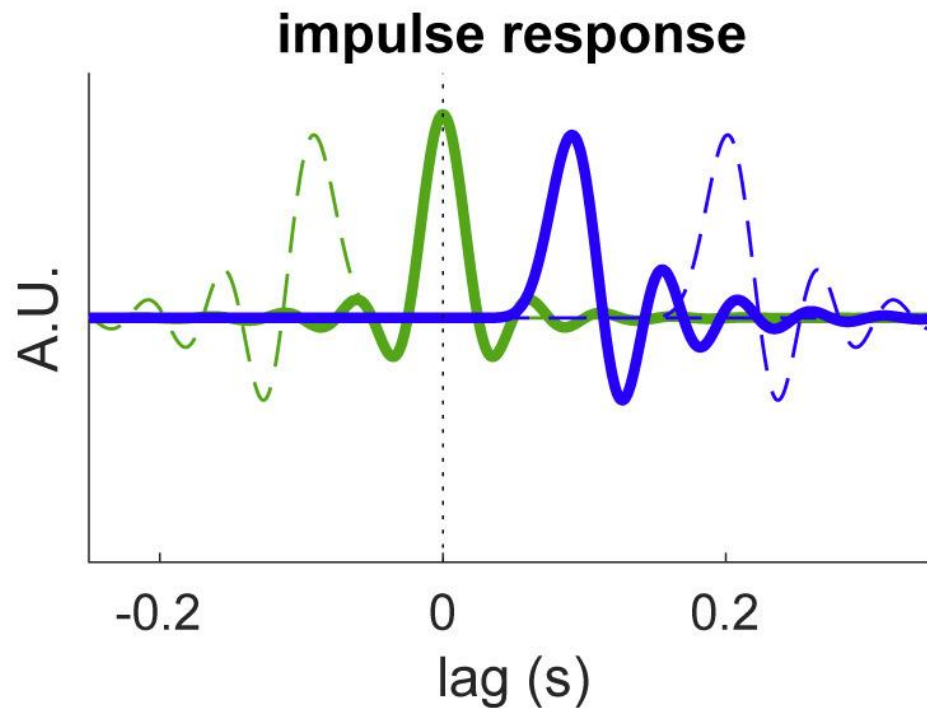


Magnitude Transfer Function and Impulse Response



Magnitude Transfer Function

Magnitude Transfer Function is not enough to fully specify the filters characteristic, you need phase information also to fully specify a filter!



Causal filter limitation

Causal filters introduce phase distortion:

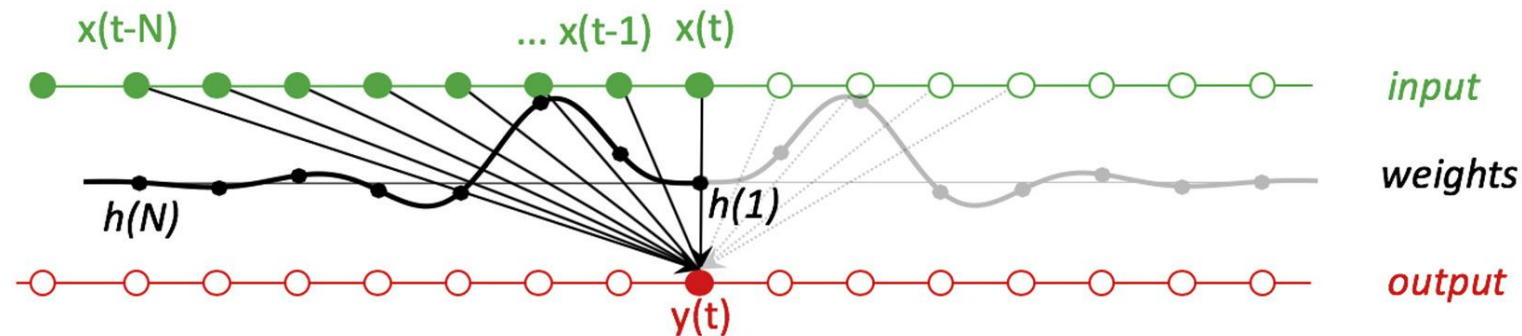
- Timing can get smeared/shifted
- Sharp features can appear to start later or be smeared asymmetrically.

Common solution: acausal filters.

- Most commonly implemented via forward-backward filtering (e.g. `filtfilt`).

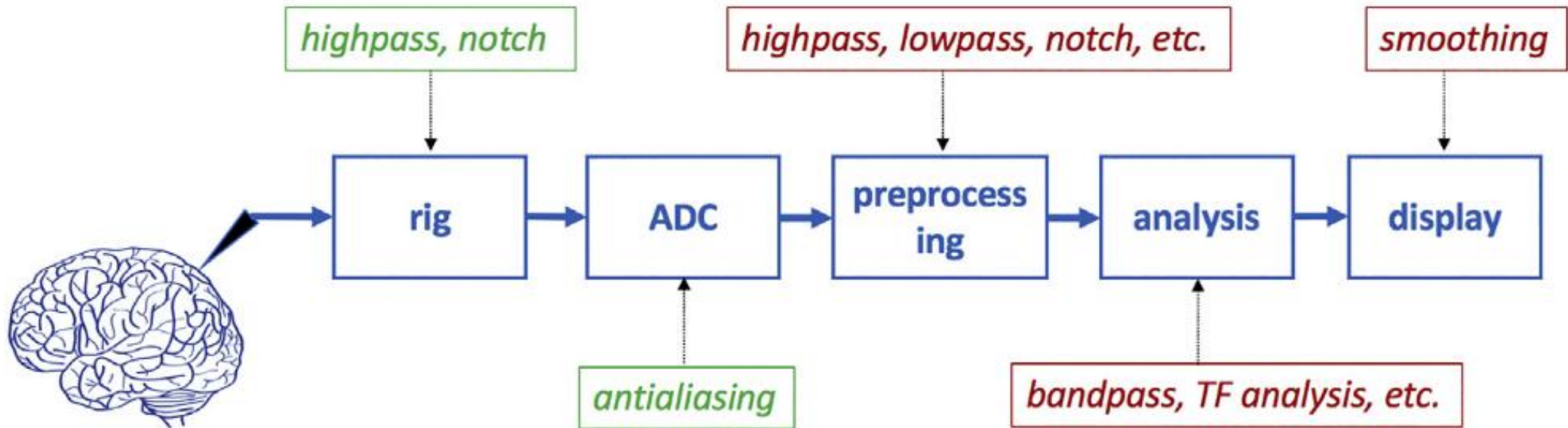
Acausal filter limitation

- Acausal filter cannot represent a physical system:
 - Physical systems are causal => the future cannot influence the past.
- However, for **offline data analysis** we can take samples from anywhere:
 - Acausal filters are realizable.
- The impulse response of zero-phase filters is symmetrical relative to zero:



- While acausal filters are easy to apply, interpreting their output requires special care!

Filters comes in many shapes and are ubiquitous



Application of filters: antialiasing (low-pass filter)

During analog-to-digital conversion or downsampling, only values at the sampling points are retained by the sampling process. Thus, the digital representation is **ambiguous**: the same set of numbers might conceivably reflect a different raw signal.

But not under certain condition!

If the original signal's spectrum contains **no power beyond the Nyquist frequency** (i.e. one half the sampling rate) then the original signal can be perfectly reconstructed from the samples.

=> Low-pass filter your data.

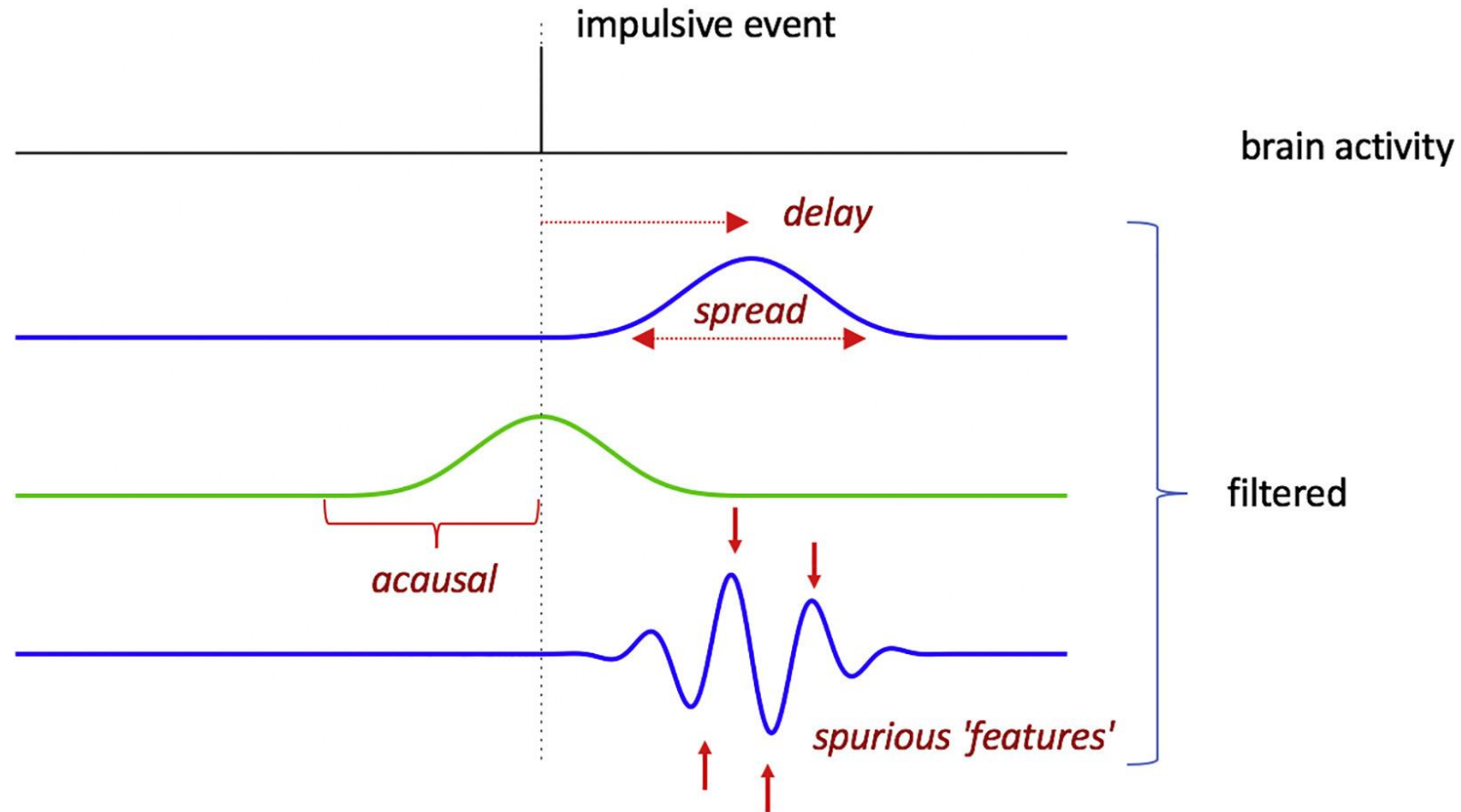
Applications of filters

- **Low-pass filters:** antialiasing, smoothing, remove high-frequency (EMG contamination)
- **High-pass filters:** remove trends and slow drifts
- **Notch filters:** remove powerline noise 50/60 Hz
- **Band-pass filters:** investigate specific frequency band (alpha, beta, ...)
- **Time-frequency analysis:** a TF representation can be viewed as the time-varying magnitude of the data at the outputs of a filterbank.

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Filter impacts on Impulse or Spike 1/2



The nature and extent of these effects depends on the filter and can be judged by looking at its impulse response.

Filter impacts on Impulse or Spike 2/2

A: Low-pass ($c_f = 10$ Hz)

B: Same, applied using `filtfilt`.

C: Butterworth low-pass ($c_f = 10$ Hz, order = 8).

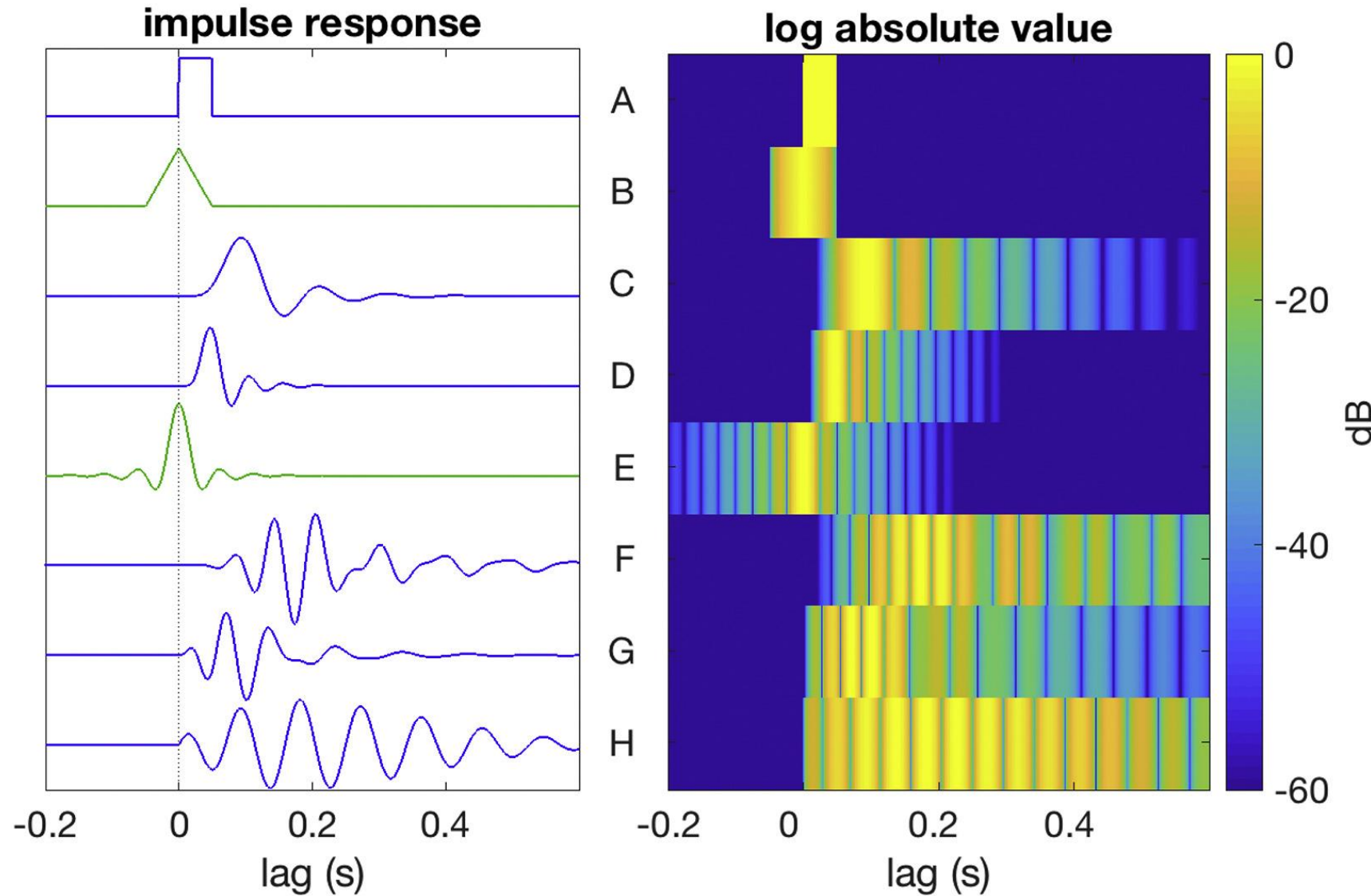
D: Same, but $c_f = 20$ Hz.

E: Same, but applied using `filtfilt`.

F: Butterworth BP 10-20 Hz, order 8.

G: Same, but order 2.

H: Same, but 10-12 Hz.



Filter impacts on Impulse or Spike 2/2

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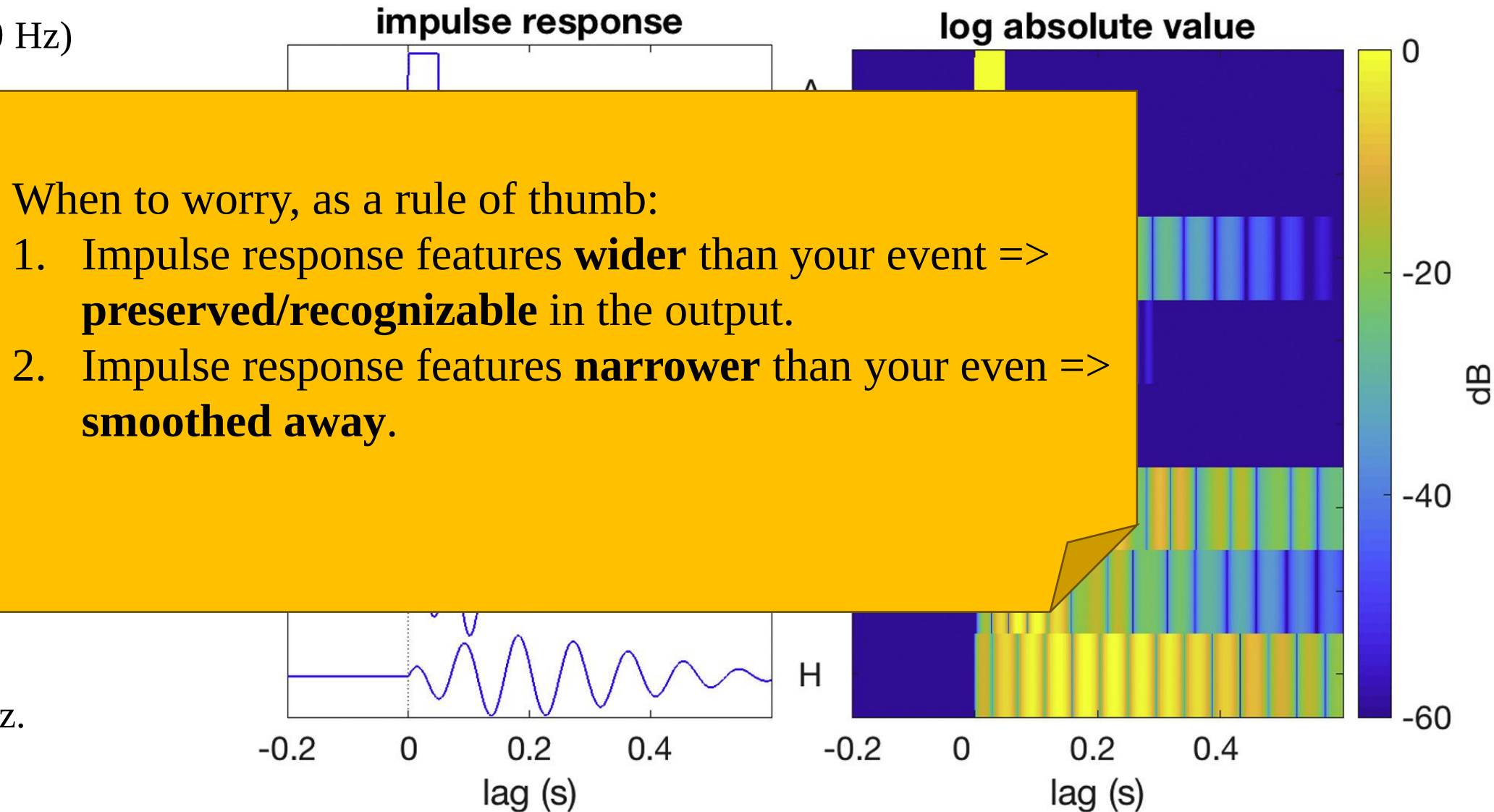
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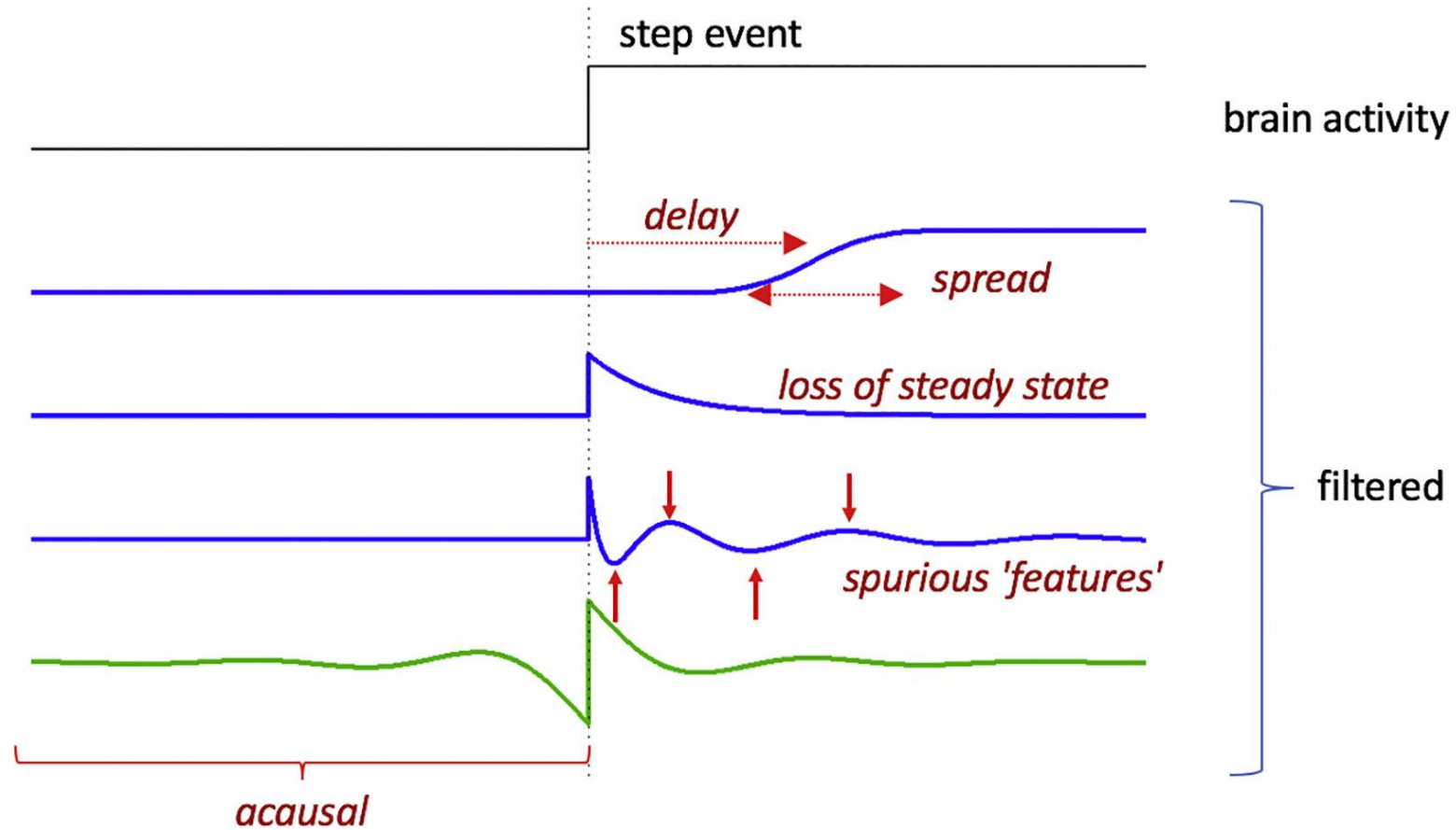
F: Butterworth BP 10-

G: Same, but order 2.

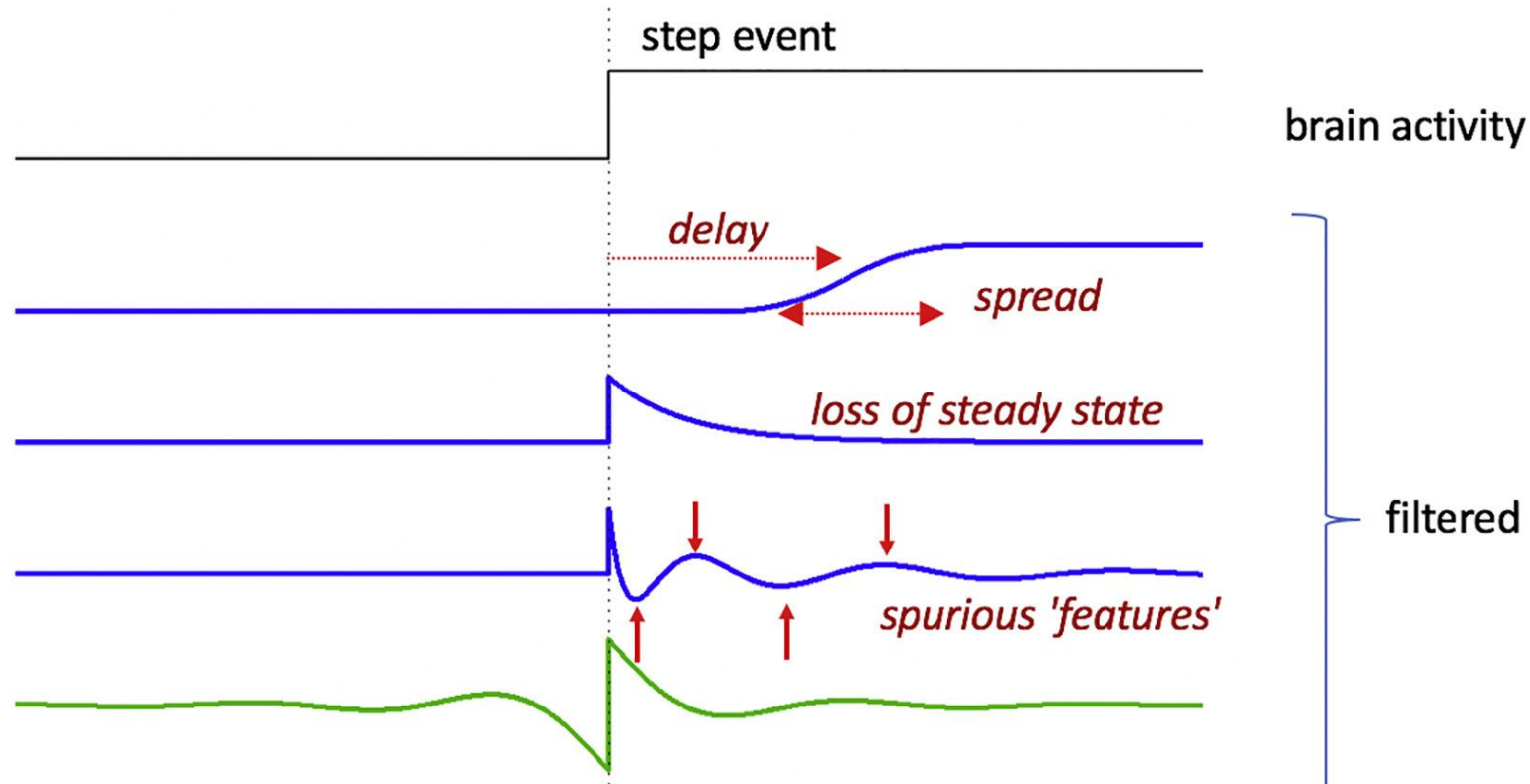
H: Same, but 10-12 Hz.



Filter impacts on step-like event 1/2



Filter impacts on step-like event 1/2



The nature and extent of these effects depends on the filter and can be judged by looking at its step response, i.e. integration over time of its impulse response.

Filter impacts on step-like event 2/2

A: Butterworth low-pass ($c_f = 10$ Hz, order 8)

B: Same, applied using filtfilt.

C: Butterworth high-pass ($c_f = 10$ Hz, order = 2).

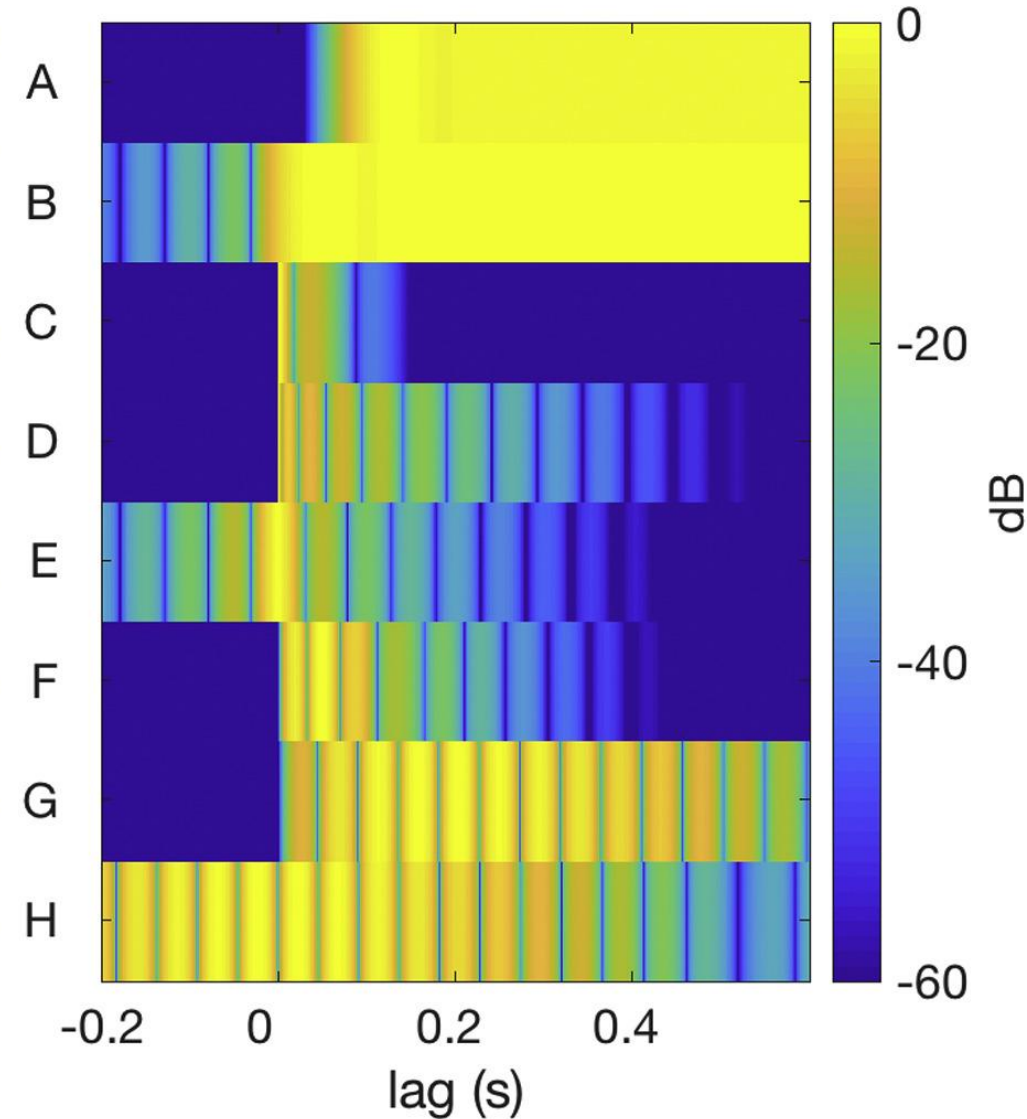
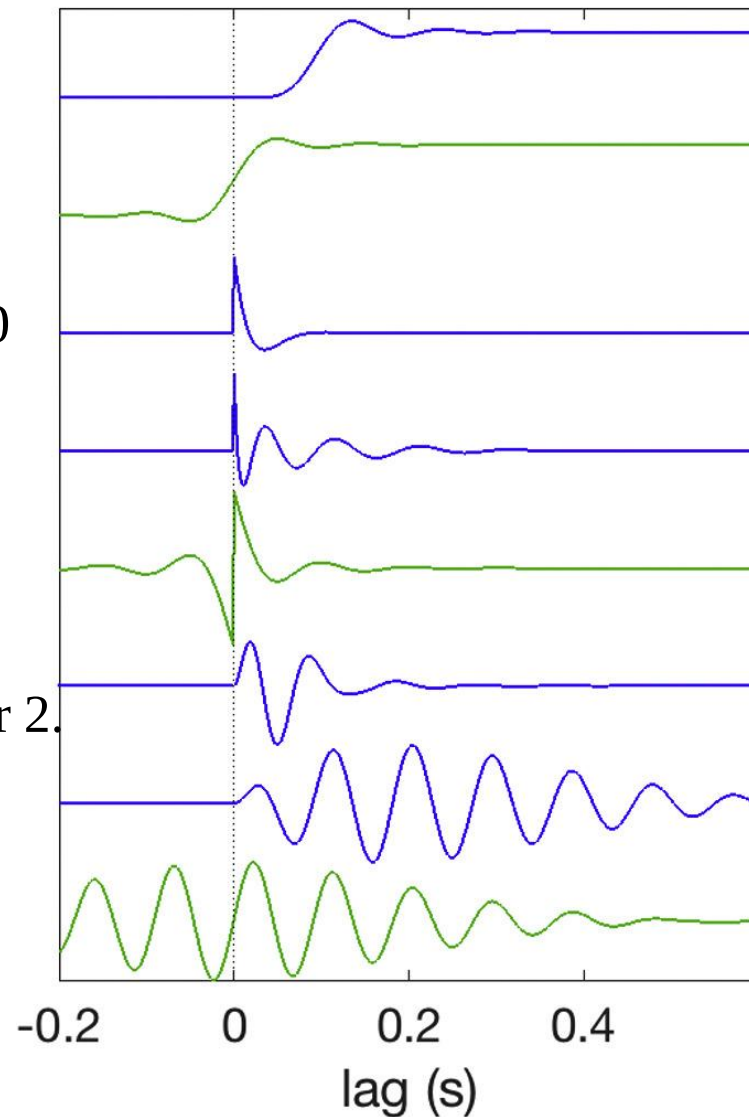
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Filter impacts on step-like event 2/2

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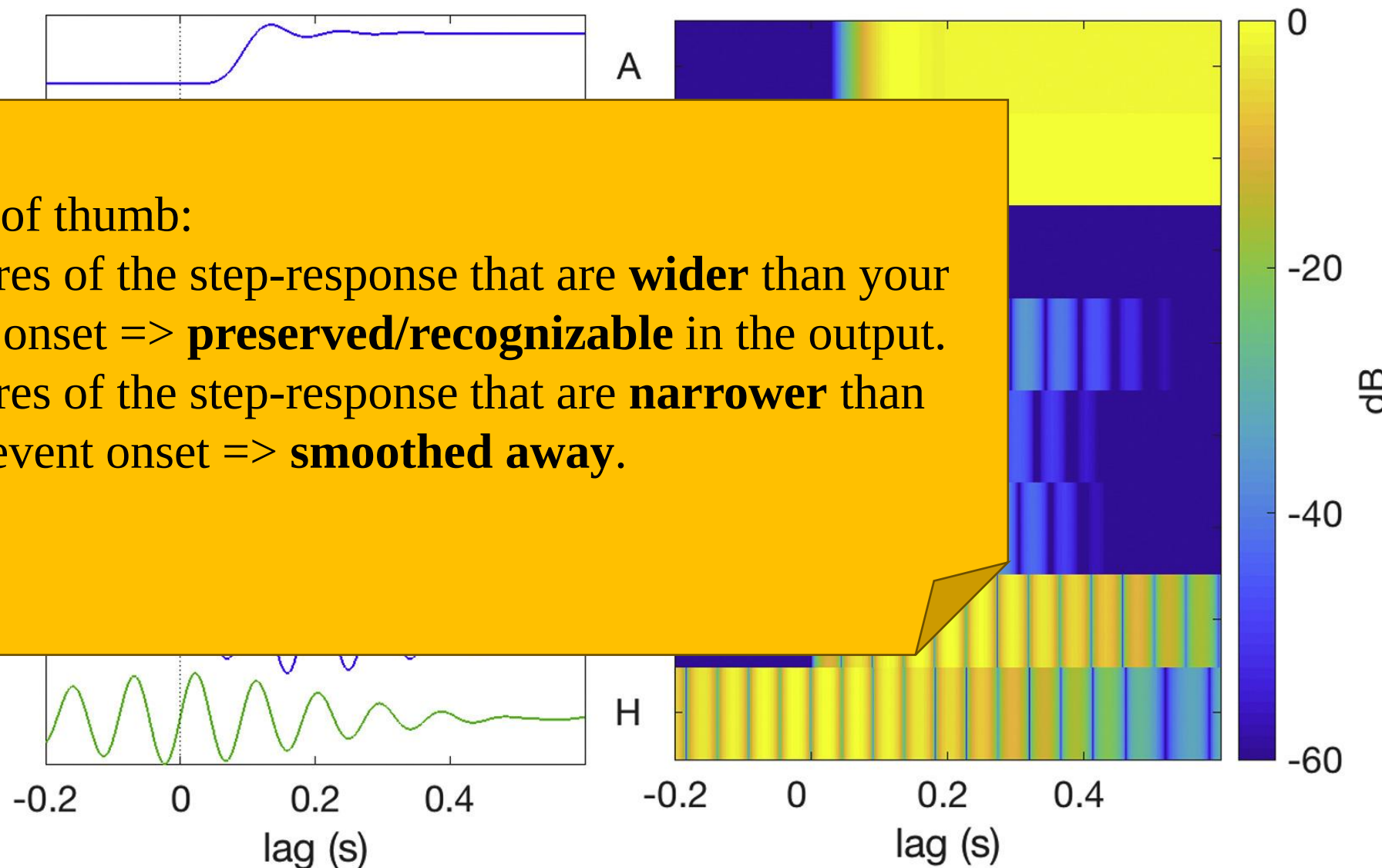
F: Butterworth BP 10-12 Hz

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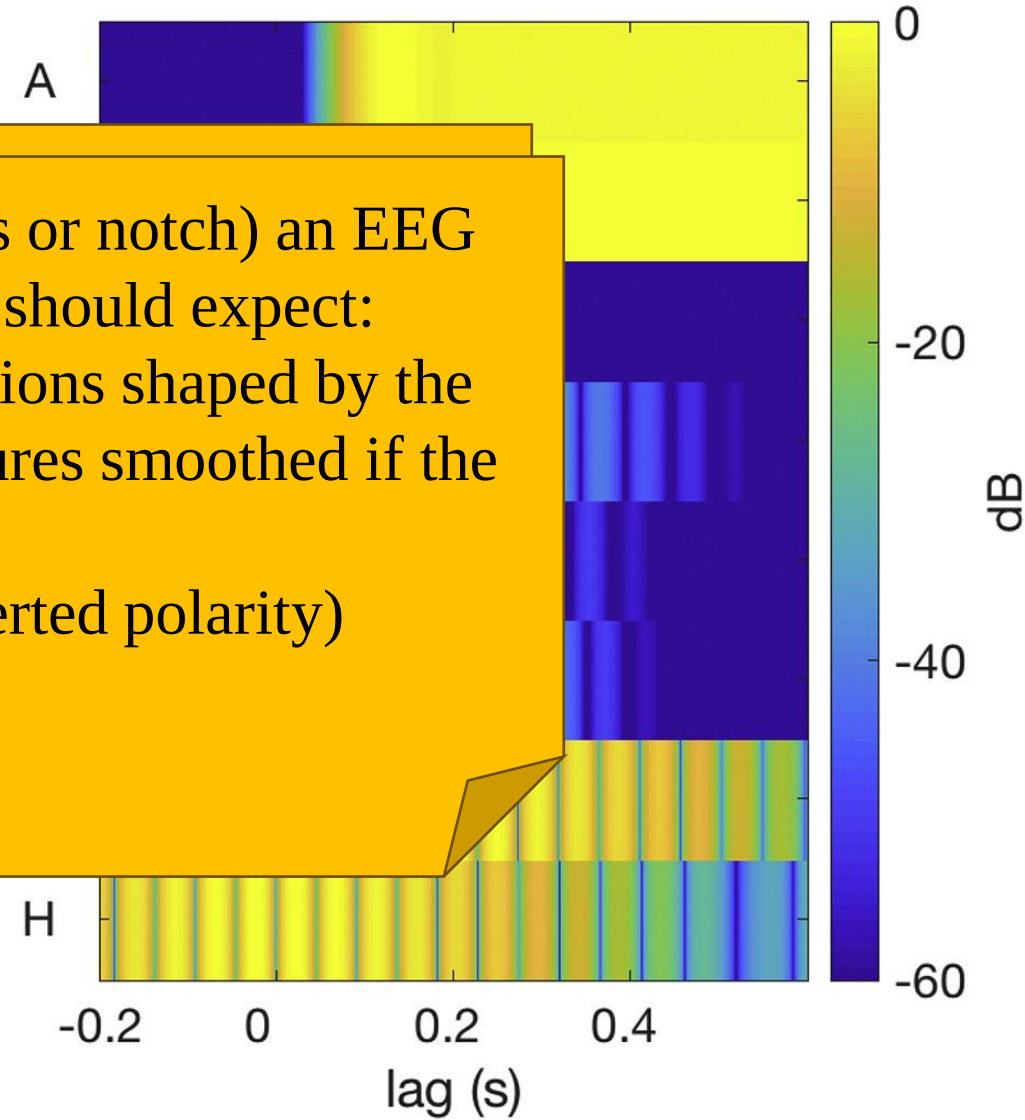
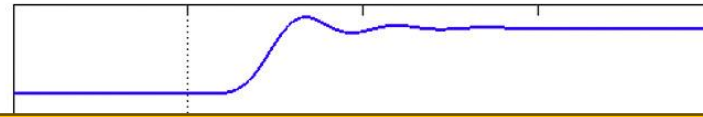
As a rule of thumb:

1. Features of the step-response that are **wider** than your event onset => **preserved/recognizable** in the output.
2. Features of the step-response that are **narrower** than your event onset => **smoothed away**.



Filter impacts on step-like event 2/2

A: Butterworth low-pass ($c_f = 10$ Hz, order 8)



B: Same, applied using

C: Butterworth high-pass (10 Hz, order = 2).

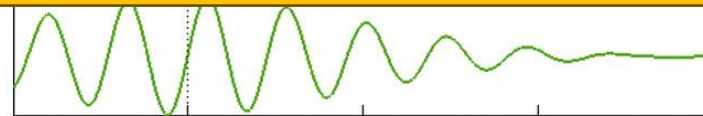
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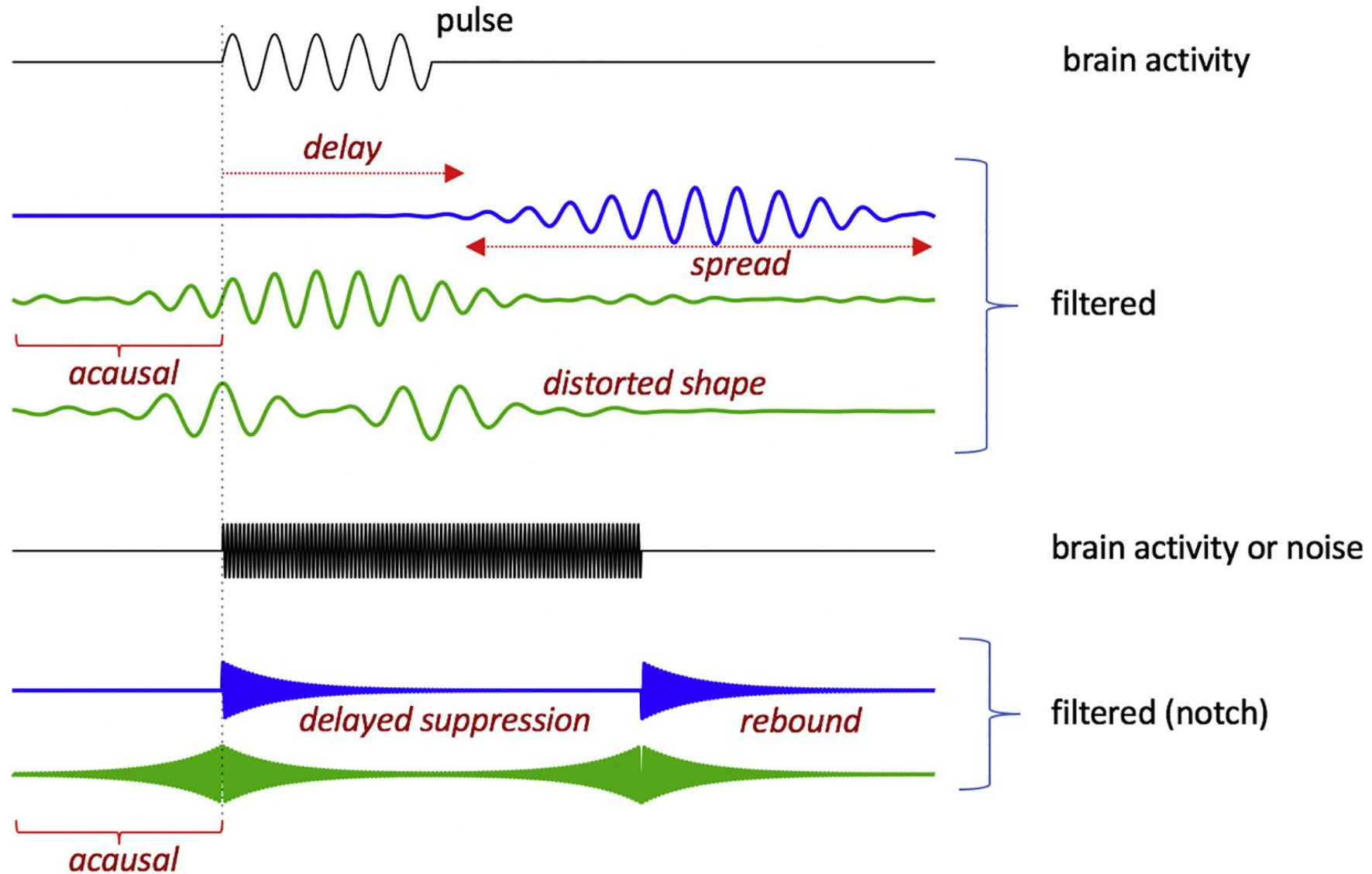


H

In practice, if you filter (e.g., high-pass or notch) an EEG signal containing a sustained shift you should expect:

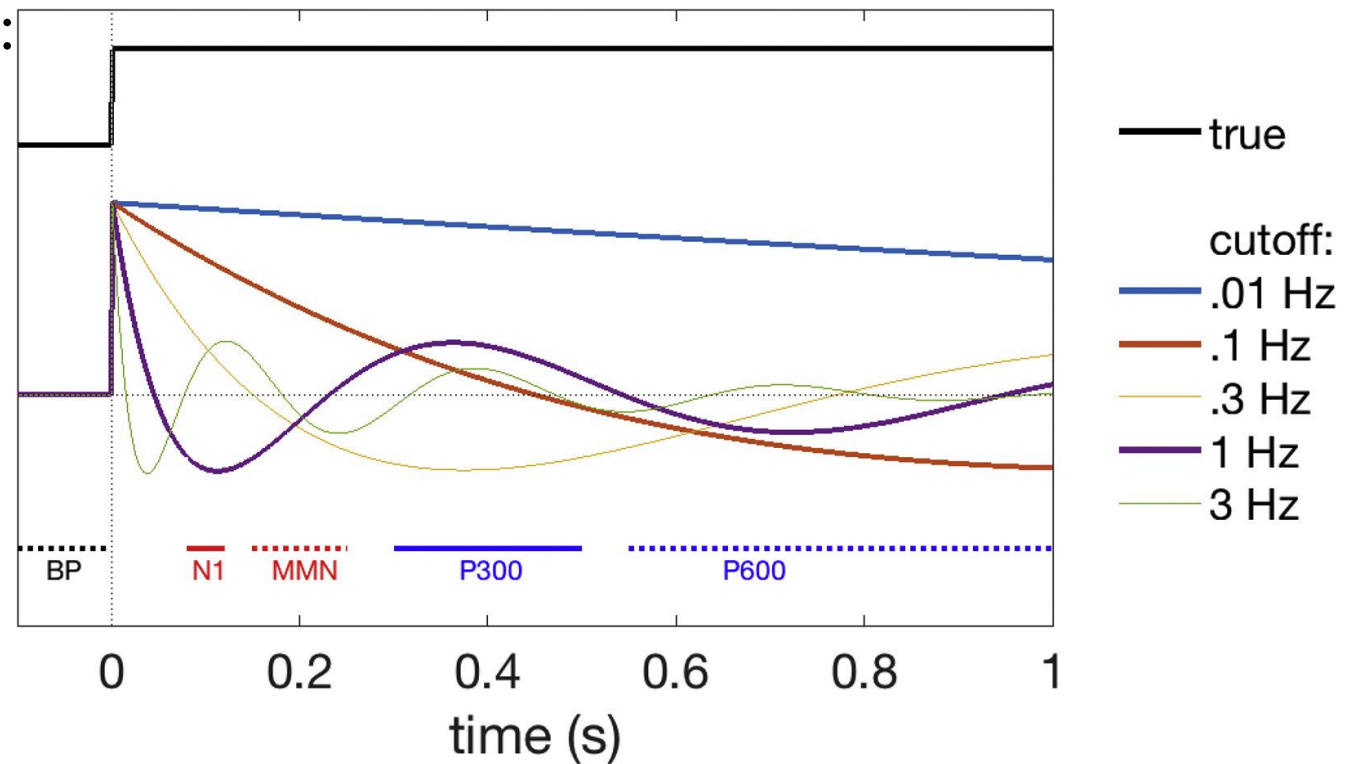
- 1. At the onset:** filter artifacts/distortions shaped by the step response, with fine/sharp features smoothed if the true onset isn't perfectly sharp.
- 2. At the offset:** a mirror-image (inverted polarity) version of the same artifact.

Filter impacts on oscillatory activity



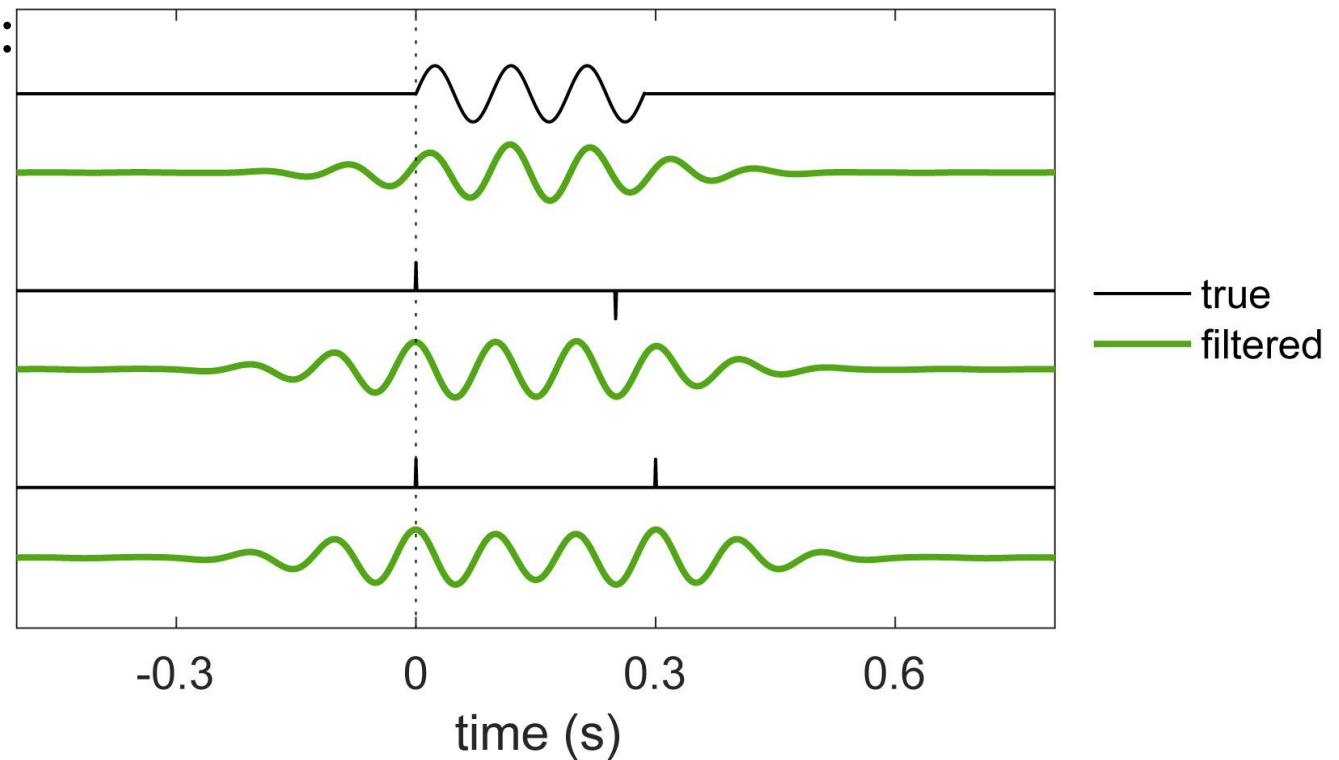
So, what can go wrong when filtering?

1. Loss of information together with the noise.
2. Distortion of the temporal features of the target:
 - peaks or transitions may be smoothed;
 - steps may turn into pulses;
 - artifactual features may appear:
 - Spurious ERP



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 - Spurious oscillations



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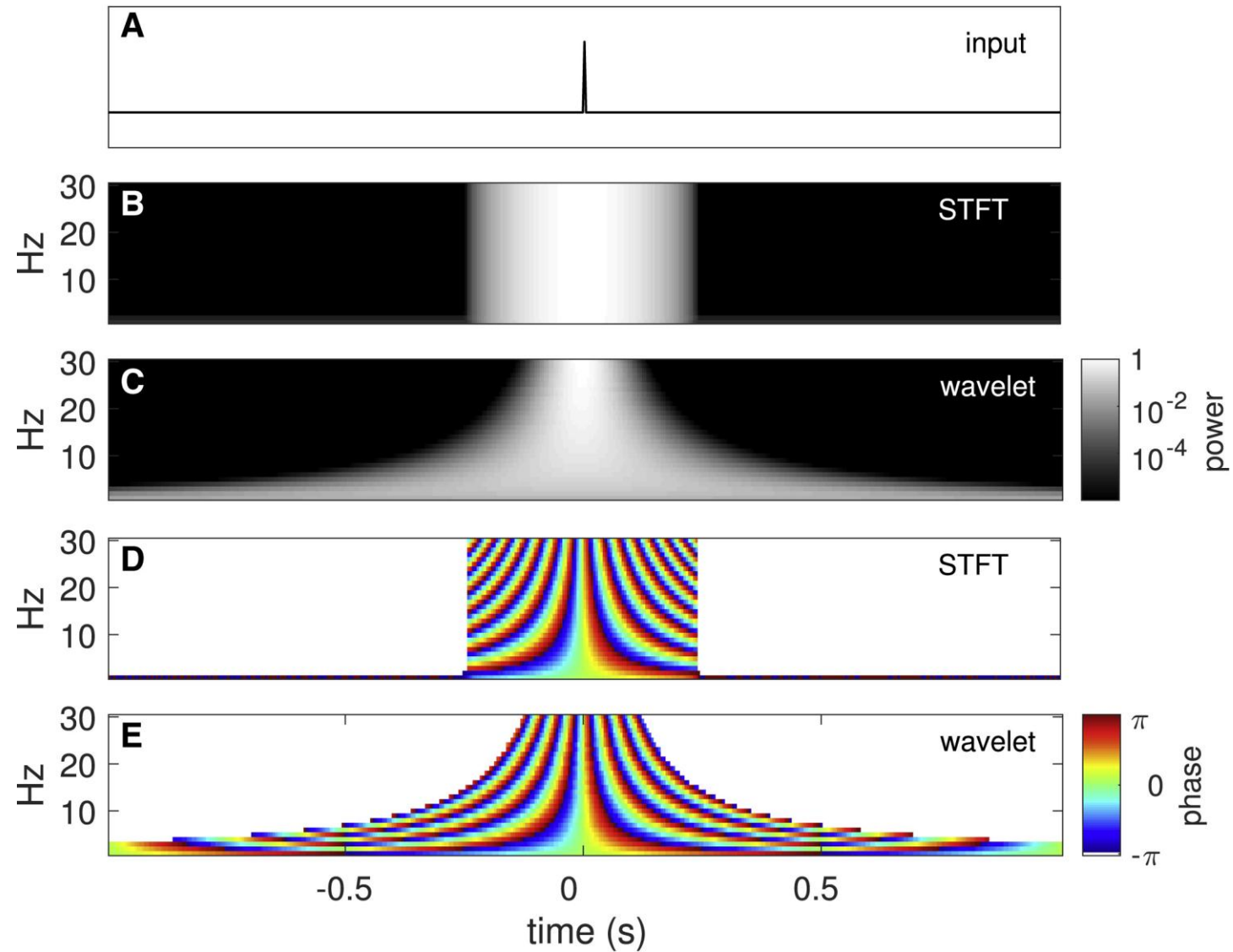
1. Loss of information together with the noise.
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 - peaks or transitions may be smoothed;
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3. Blurring of temporal relationship or causal relations:
 - Interfere with the comparison between brain measurements and stimulation or behavior;
 - Interfere between recordings at different recording sites;
 - Interfere between different frequency bands.

The problem is mild if the filter impulse response is short relative to the phenomena being measured.

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4. Problems naturally arise in TF analysis too.

So, what can go wrong when filtering? TF Analysis



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- Interfere betw
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Reports enough details on the filters so that the reader can understand them too.

ents and stimulation

;

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```
filt_params = dict(  
    sfreq=sfreq,  
    l_freq=l_freq,  
    h_freq=h_freq,  
    method='fir',  
    phase='minimum',  
    fir_design='firwin',  
    verbose=True  
)
```

Setting up band-pass filter from 8 - 12 Hz

FIR filter parameters

Designing a one-pass, non-linear phase, causal bandpass filter:

- Windowed time-domain design (firwin) method
- Hamming window with 0.0194 passband ripple and 53 dB stopband attenuation
- Lower transition bandwidth: 2.00 Hz
- Upper transition bandwidth: 3.00 Hz
- Filter length: 423 samples (1.652 s)

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Let's measure and define “short”:

For causal filters...

Calculate the 95% effective impulse response length of your filter(s):

1. Use synthetic signal to make a singular impulse;
2. Apply your filter(s);
3. Square the obtain impulse response;
4. Calculate the cumulative sum;
5. Find the timepoint matching 95% of the maximum value.

That's the minimum delay needed by your filter(s) to actually converge. By **especially cautious** in your interpretation for whatever you find before that delay or if your inter-stimulus is shorter than that.

For acausal filters, simply do that in both direction separately.

Let's measure and define “short”:

For causal filters...

Calculate the 95% effective impulse response length of your filter(s):

1. Use synthetic signal to make a singular impulse;
2. Apply your filter(s);
3. Square the obtain impulse response;
4. Calculate the cumulative sum;
5. Find the

This small code will be available by the end of the week on my github:
<https://github.com/Jimmy-Petit>

Especially cautious in your interpretation
for whatever you find before that delay or
if your inter-stimulus is shorter than that.

For acausal filters,
simply do that in
both direction
separately.

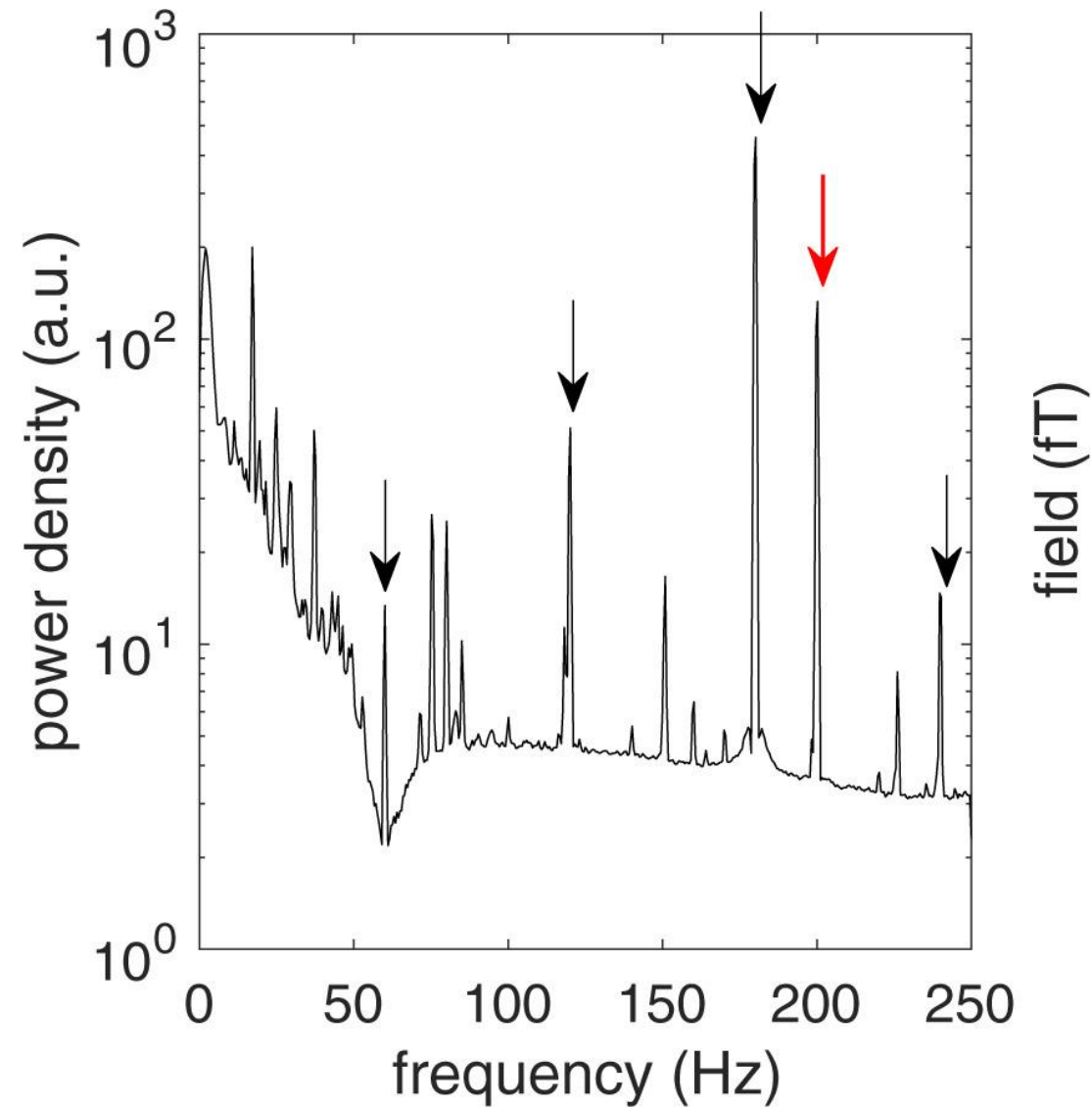
Plan

- Definition and Motivation:
 - What's a signal or noise?
 - Noise in EEG
- What's a filter?
 - Definition
 - Intuition behind dot product and convolution: the fondation of filtering!
 - Causal and Acausal filters limitations
- Danger zone
 - Impact of filters on standard signal (rule of thumbs on what to expect)
 - The hidden costs: phase distortion, edge artifacts, temporal smearing, spurious oscilliations, etc.
- How to fix it?
 - Low-pass, high-pass, band-pass, etc.
- Conclusion and Practical guidelines

How to fix these issues?

Reports enough details on the filters so that the reader can understand them too.

How to fix these issues? Antialiasing – analog-to-digital



How to fix these issues? Antialiasing – digital-to-digital

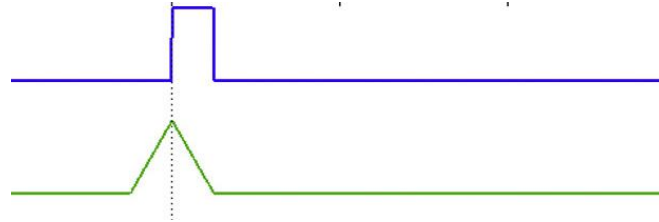
If you downsample or resample your data, two possible solution:

1. Apply an antialiasing filter before downsampling. To reduce temporal smear and ringing, consider using an antialiasing filter with a lower cutoff and shallower slope.
2. Or better: consider alternatives such as interpolation (e.g., linear, cubic, or spline)!
 - Essentially, interpolation acts as a gentle, low-distortion digital antialiasing filter that is applied as part of the resampling process itself.
 - More info: *Robust detrending, rereferencing, outlier detection, and inpainting for multichannel data*, **Alain de Cheveigné and Dorothée Arzounian**, NeuroImage, 2026.

How to fix these issues? Low-pass filtering

What's the aim?

1. smooth the temporal waveform => consider using simple kernel:



2. to ensure attenuation of high-frequency power:
 - same as for antialiasing digital-to-digital;
 - spatial filters may be applied to remove noise sources (Parra et al., 2005; de Cheveigné and Parra, 2014).

How to fix these issues? High-pass filtering

What's the aim?

1. Remove a DC drift?
 - Consider substrating the overall mean
2. Removing a slow trend?
 - consider “detrending” rather than high-pass filtering (de Cheveigné and Arzounian, 2018)
 - consider “ regression” algorithms too (Vrba and Robinson, 2001; de Cheveigné and Simon, 2007)
3. If all else fails, then use a high-pass filter, but:
 - pay attention to its morphology of responses;
 - if the initial portion of the data is on average far from zero: subtract the average over that portion as to limit edge artifact;
 - if the data are to be cut into, it is usually best to filter the continuous data first.

How to fix these issues? Band-pass filtering

Nothing much can be done here.

1. Consider whether a band-pass filter is really needed.
2. Consider filters with relatively shallow slopes, and cutoff frequencies distant from the activity of interest.
3. Be on the lookout for artifactual results due to the filtering.

How to fix these issues? Notch filtering

What's the aim?

1. Suppress power-line noise?
 - Consider remove the noise at the source and have everything powered on batterie.
 - When not possible: measure the line noise on one reference channel and regress it out (Vrba and Robinson, 2001).
 - If the data are multichannel, consider using component analysis to isolate the line noise components and regress them out.
2. If the high-frequency region is not of interest:
 - Boxcar smoothing kernel of size 1/50 Hz (or 1/60 Hz as appropriate).
 - However: if sampling frequency isn't a multiple of the powerline, you can't have exactly the number of samples in the boxcar. Fix that using interpolation! (de Cheveigné and Arzounian, 2018).

How to fix these issues? TF Analysis

1. If the pattern of interest can be interpreted in the time domain:
=> avoid TF analysis.
2. Otherwise:
 - consider using fixed kernelsize analysis (e.g., DFT) rather than, or in addition to, wavelet analysis, so that temporal bias and smearing are uniform across the frequency axis.
 - consider using relatively short analysis windows to reduce temporal bias and/or smearing.
 - weigh carefully the choice between causal analysis versus acausal analysis:
 - Temporal bias but no causality issues;
 - No temporal bias but risk of misleading causal relations.

How to fix these issues? TF Analysis

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 - consider using relatively short analysis windows to reduce temporal bias and/or smearing
 - weigh carefully versus acausal analysis:
 - Temporal bias
 - No temporal

One should be particularly concerned if an interesting effect only emerges with a particular analysis method.

Conclusion - Potential issue coming from filtering:

1. Loss of information in spectral regions suppressed by the filter.
2. Distortion of response waveforms and the emergence of spurious features.
3. Blurring of temporal relations: violation of causality.
4. Same event can be represented by different shape: depending on the analysis.

Conclusion - Guidelines for better practices:

1. Know your filters and document more details on the filters and pipeline.
2. Any filter entails scrambling of the temporal axis.
3. Be aware of acausal processing (e.g. filter or TF centered on time-window).
4. Know your noise and eliminate it at the source when possible:
 - Line power noise: banish power cables from the vicinity of the setup, use lights fed with filtered DC, apply proper shielding, avoid cables loops, avoid GND loop, etc.
 - High-frequency noise: banish computer screens, fluorescent lights, cell phones, etc.
 - To minimize slow drifts in EEG: proper electrode application and keep the subjects cool.
 - To minimize alpha components: ensure that subjects keep their eyes open and give them a task to keep them alert.

Conclusion - Guidelines for better practices:

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2. Any filter entails scrambling of the temporal axis.
3. Be aware of acausal processing (e.g. filter or TF centered on time-window).
4. Know your noise and eliminate it at the source when possible.
5. Consider alternatives to filtering.
6. Do some simulation, especially in case of cascade filters.

Thank you for your attention!

References and ressources used for this workshop:

- Platform: <https://www.learningeeg.com/>
- Chapter 10 of *Analyzing Neural Time Series Data: Theory and Practice*, Mike X **Cohen**, 2014
- *Filters: When, Why, and How (Not) to Use Them*, Alain **de Cheveigné** and Israel **Nelken**, Neuron, 2019
- *Digital filter design for electrophysiological data – a practical approach*, Andreas **Widmann**, Erich **Schröger**, Burkhard **Maess**, J. Neurosci. Methods., 2015
- *Four common conceptual fallacies in mapping the time course of recognition*, Rufin **VanRullen**, Front. Psychol., 2011
- *Does Filtering Preclude Us from Studying ERP Time-Courses?*, Guillaume **Rousselet**, Front. Psychol., 2012